



US012417086B2

(12) **United States Patent**
Avedisov et al.

(10) **Patent No.:** **US 12,417,086 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **DATA COMMUNICATION WITH REMOTE OPERATION OF A VEHICLE**

(56) **References Cited**

U.S. PATENT DOCUMENTS

(71) Applicant: **TOYOTA MOTOR ENGINEERING & MANUFACTURING NORTH AMERICA, INC.**, Plano, TX (US)

7,447,576 B2 11/2008 Ban et al.
8,155,819 B2 4/2012 Barker
(Continued)

(72) Inventors: **Sergei S. Avedisov**, Cupertino, CA (US); **Mohammad Irfan Khan**, Mountain View, CA (US); **Onur Altintas**, Mountain View, CA (US)

FOREIGN PATENT DOCUMENTS

EP 0817734 A1 1/1998
EP 2480443 A1 8/2012
(Continued)

(73) Assignees: **TOYOTA MOTOR ENGINEERING & MANUFACTURING NORTH AMERICA, INC.**, Plano, TX (US); **TOYOTA JIDOSHA KABUSHIKI KAISHA**, Aichi-Ken (JP)

OTHER PUBLICATIONS

Baskaran et al. 5G-connected remote-controlled semi autonomous car trial. 2020 IEEE International Conference on Machine Learning and Applied Network Technologies.

(Continued)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 177 days.

Primary Examiner — S. Sough

Assistant Examiner — Mohammad H Kabir

(21) Appl. No.: **18/220,223**

(22) Filed: **Jul. 10, 2023**

(65) **Prior Publication Data**

US 2025/0021319 A1 Jan. 16, 2025

(51) **Int. Cl.**
G06F 21/30 (2013.01)
B60R 16/023 (2006.01)
(Continued)

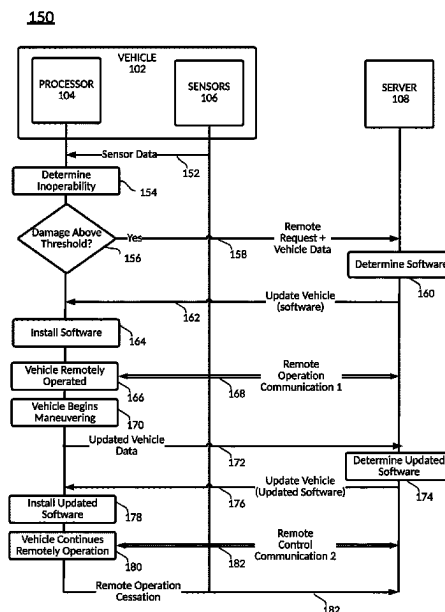
(52) **U.S. Cl.**
CPC **G06F 8/61** (2013.01); **B60R 16/0231** (2013.01); **G06F 8/453** (2013.01)

(58) **Field of Classification Search**
CPC G06F 8/61; G06F 8/453; B60R 16/0231
See application file for complete search history.

(57) **ABSTRACT**

An example operation includes one or more of responsive to an inoperability of a vehicle above a threshold, requesting for the vehicle to be remotely operated, receiving software, by the vehicle, based on the level of inoperability, responsive to an installation of the software on the vehicle, remotely operating the vehicle using the installed software, responsive to the inoperability of the vehicle above the threshold while it is remotely operated, receiving an update to the software, responsive to an installation of the updated software on the vehicle, further remotely operating the vehicle using the installed updated software, and responsive to the inoperability below the threshold, requesting a cessation of the further remote operation of the vehicle.

17 Claims, 28 Drawing Sheets



- * cited by examiner

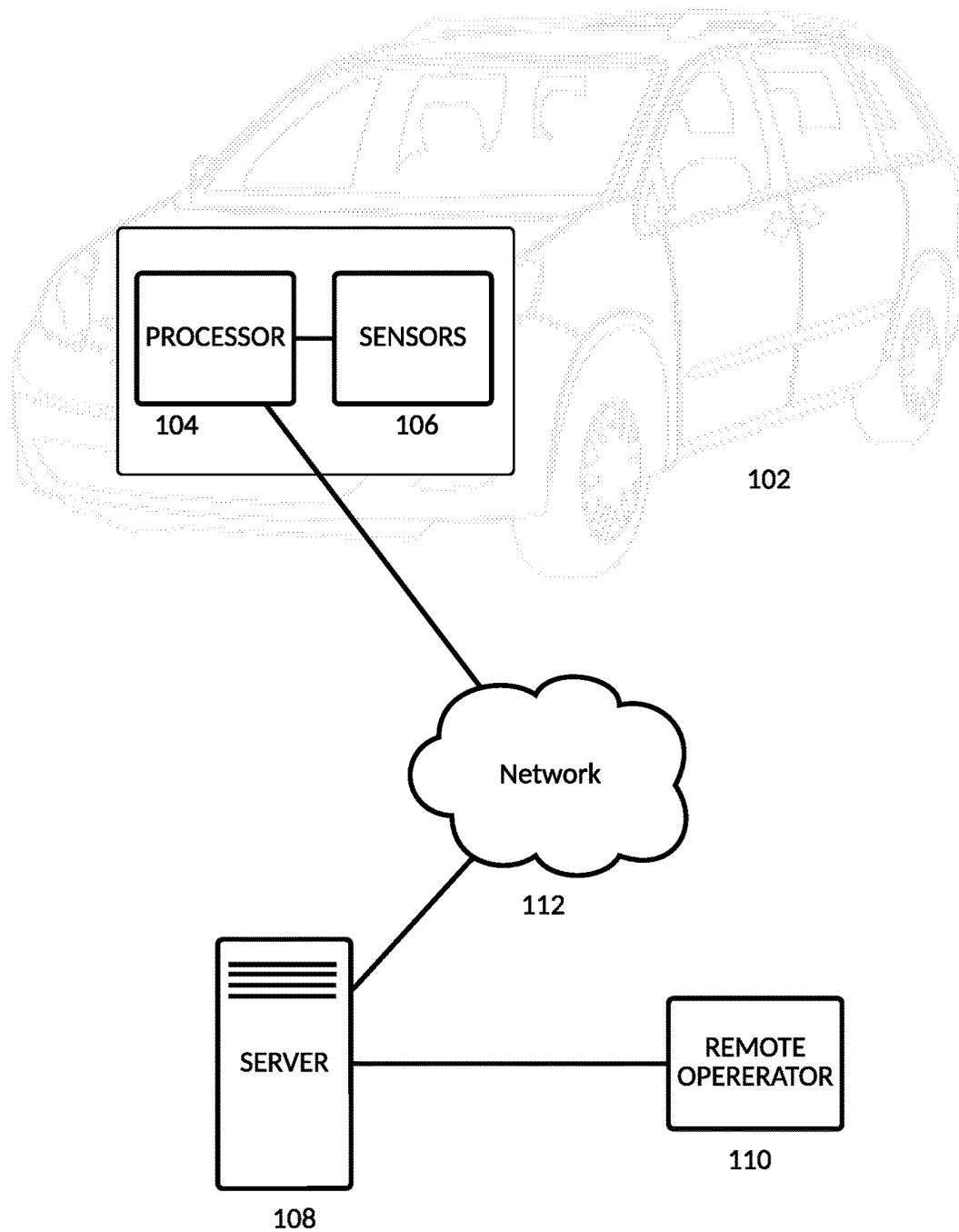
100

FIG. 1A

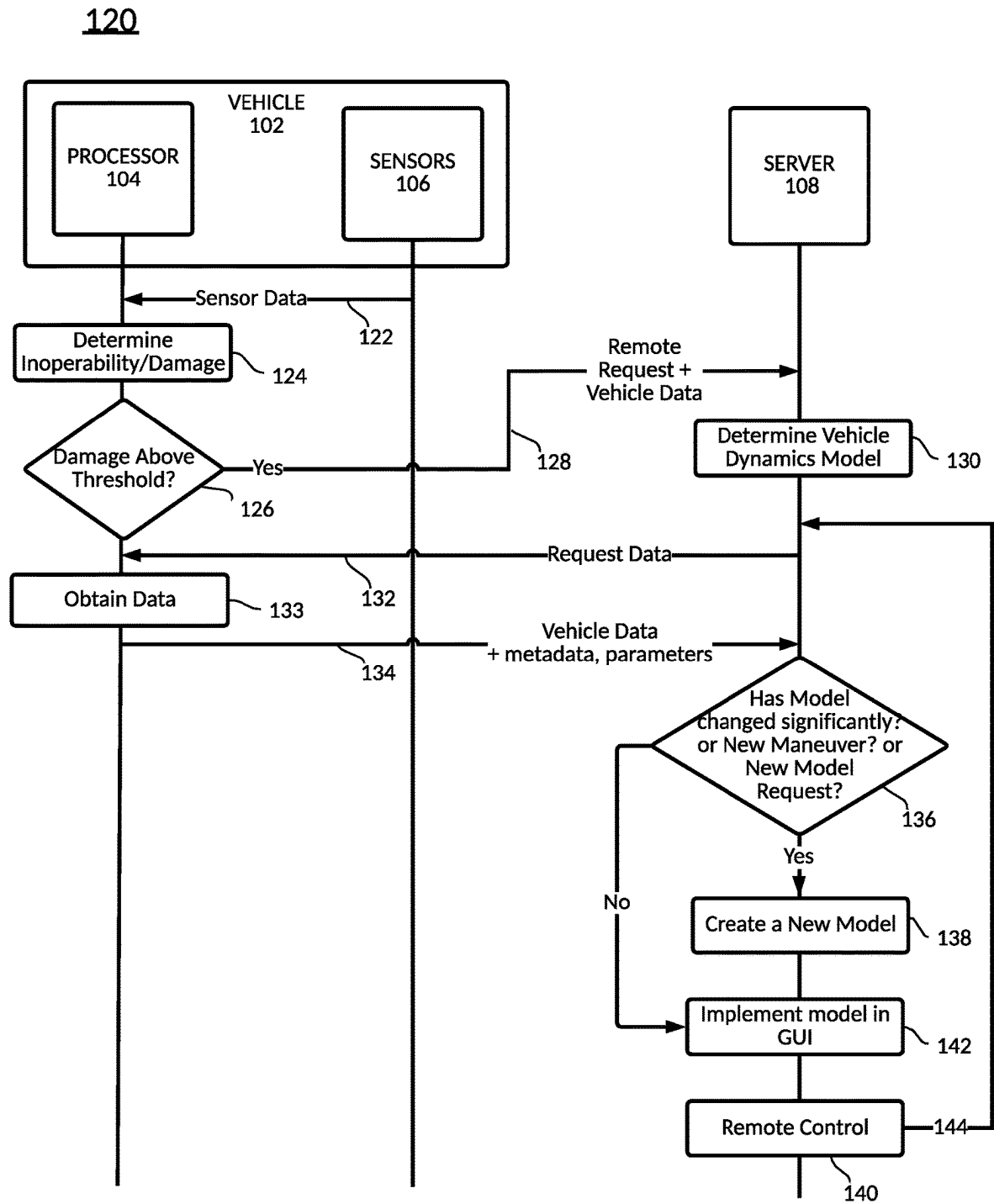


FIG. 1B

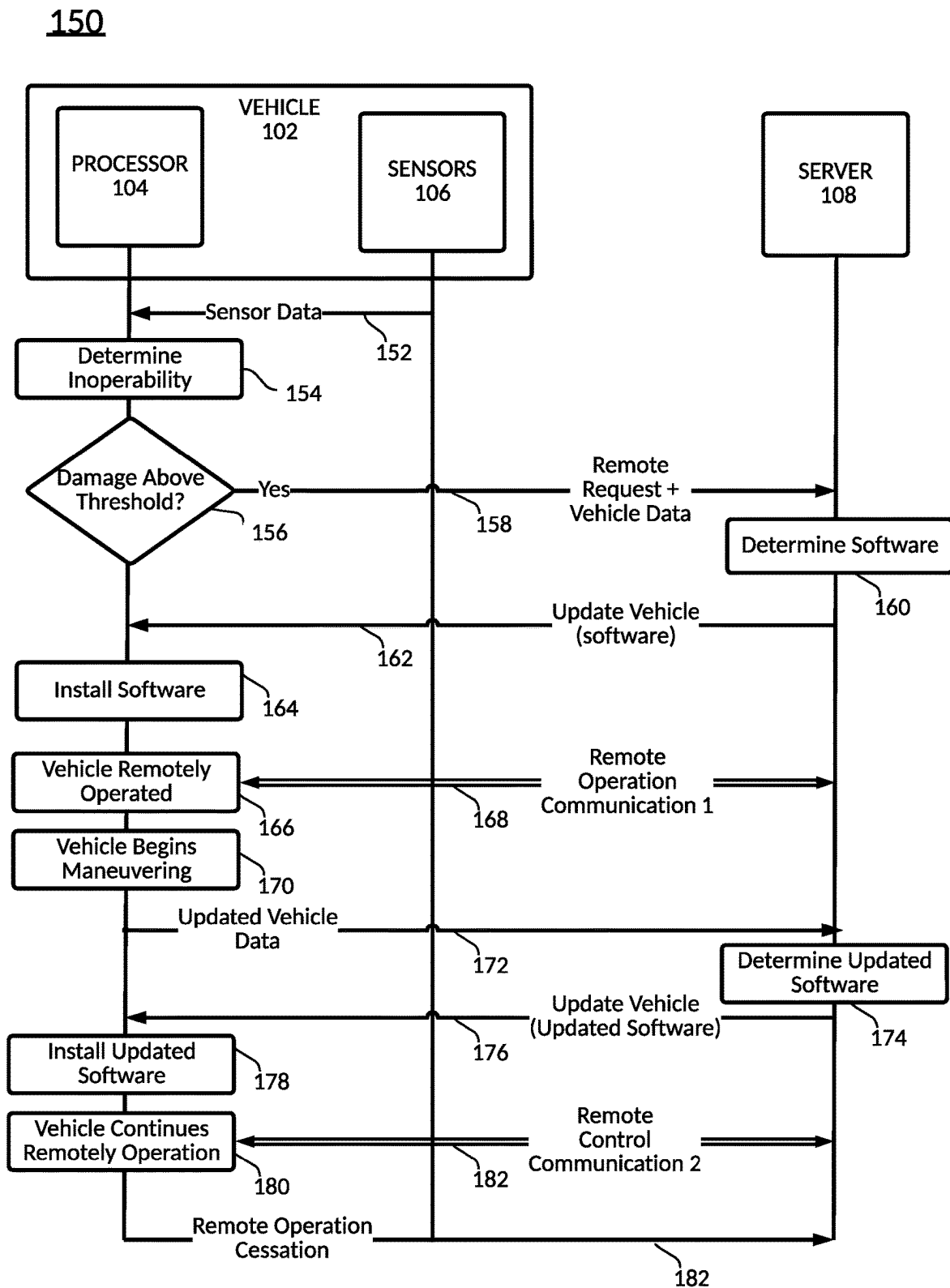


FIG. 1C

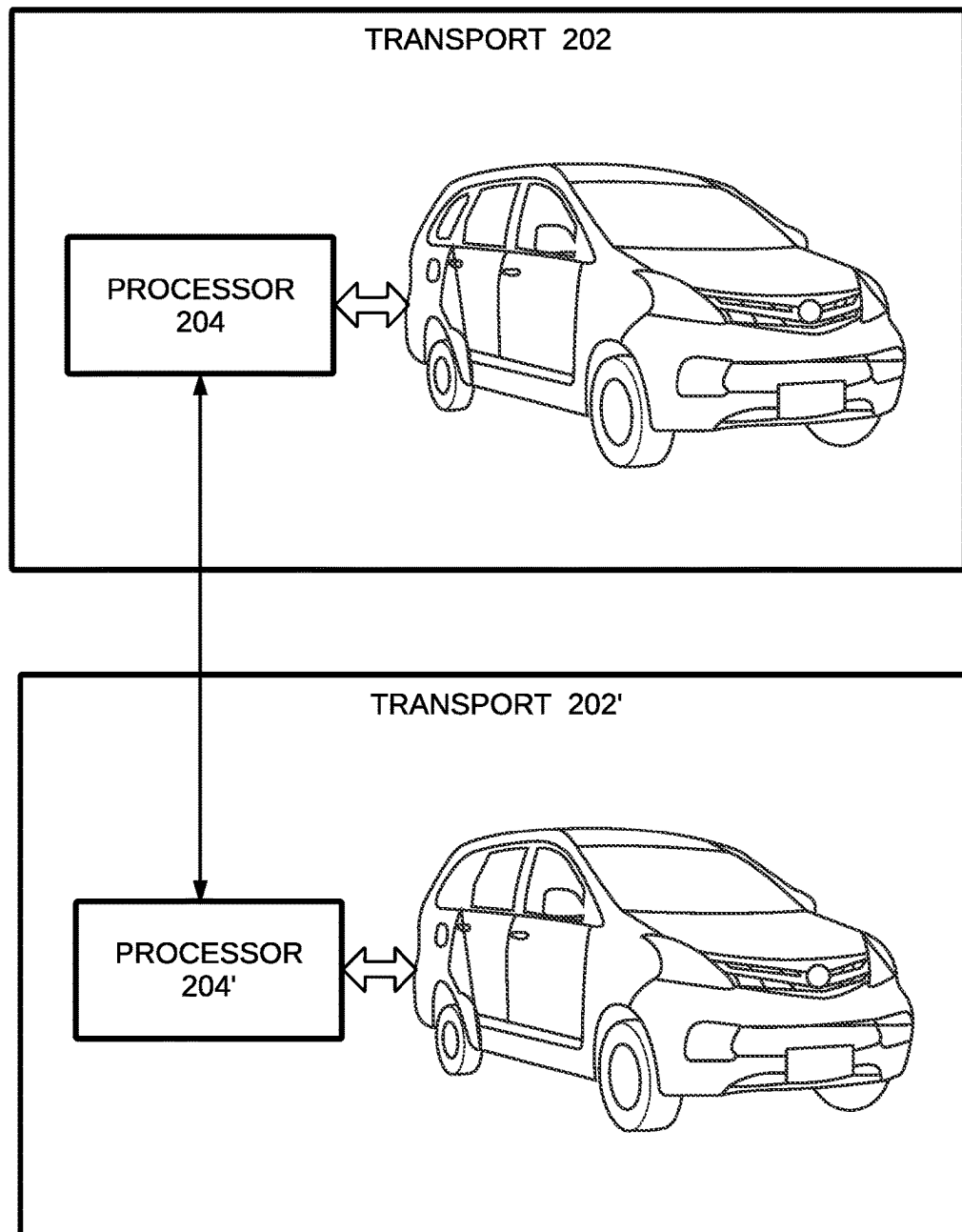
200

FIG. 2A

210

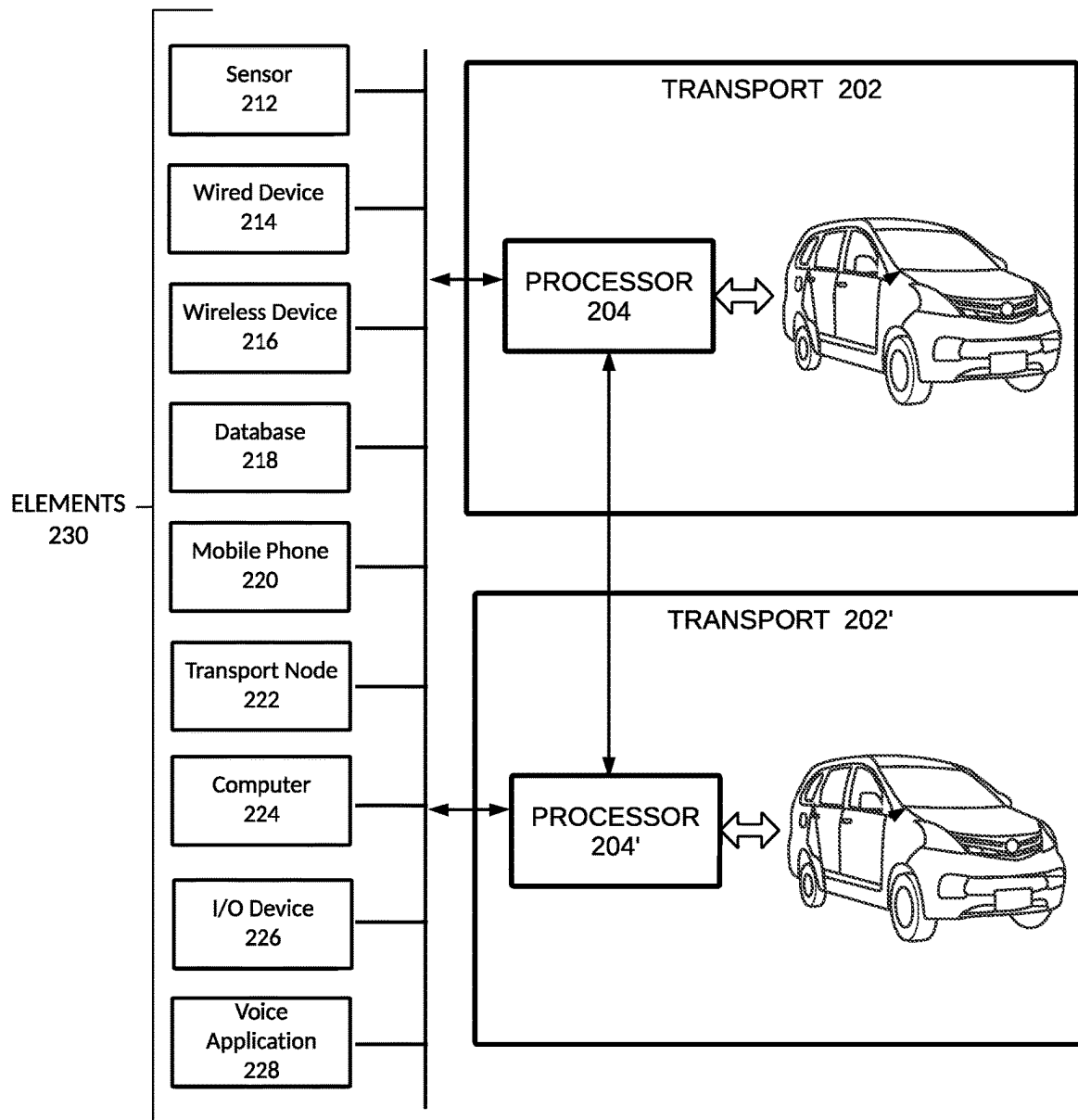


FIG. 2B

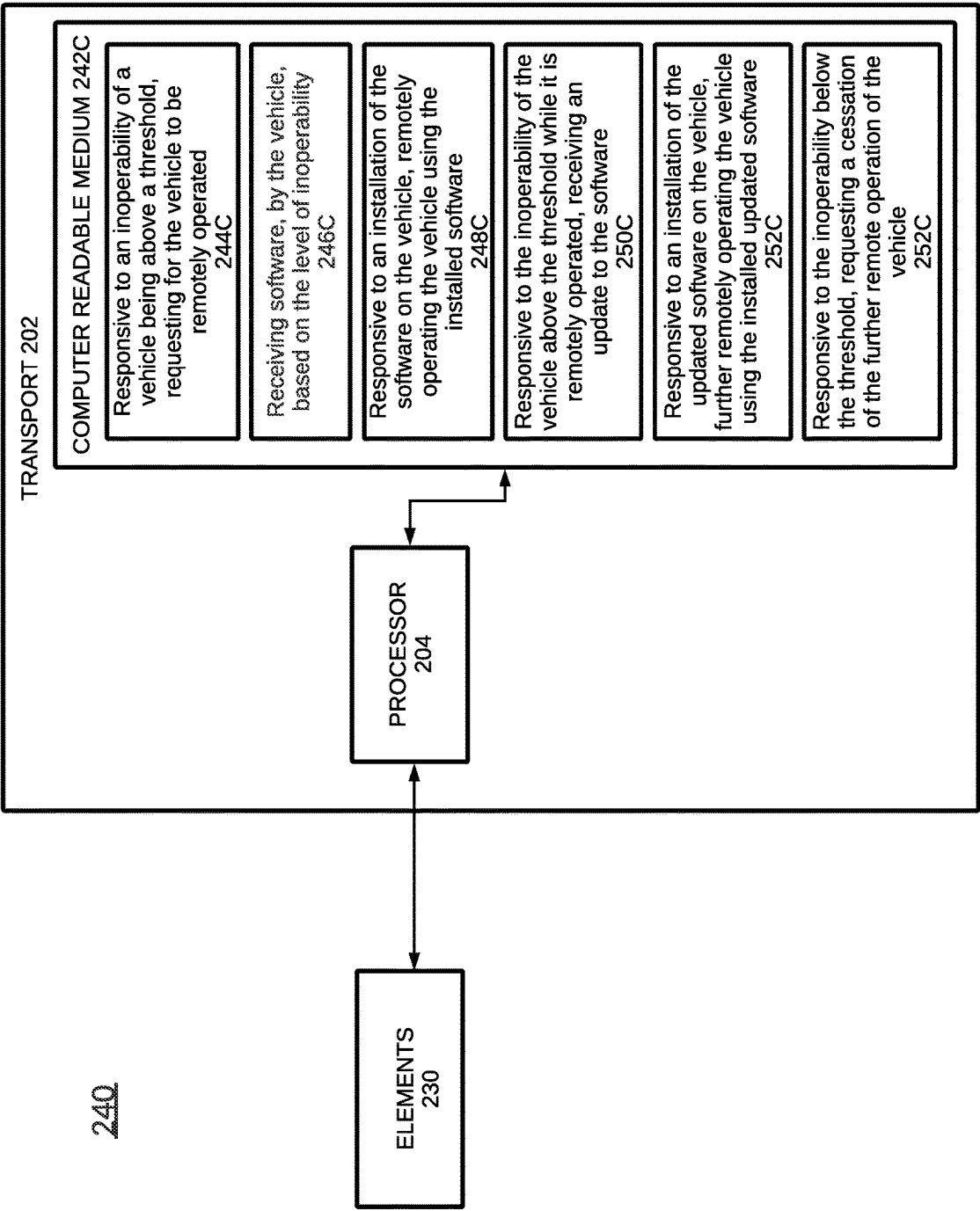


FIG. 2C

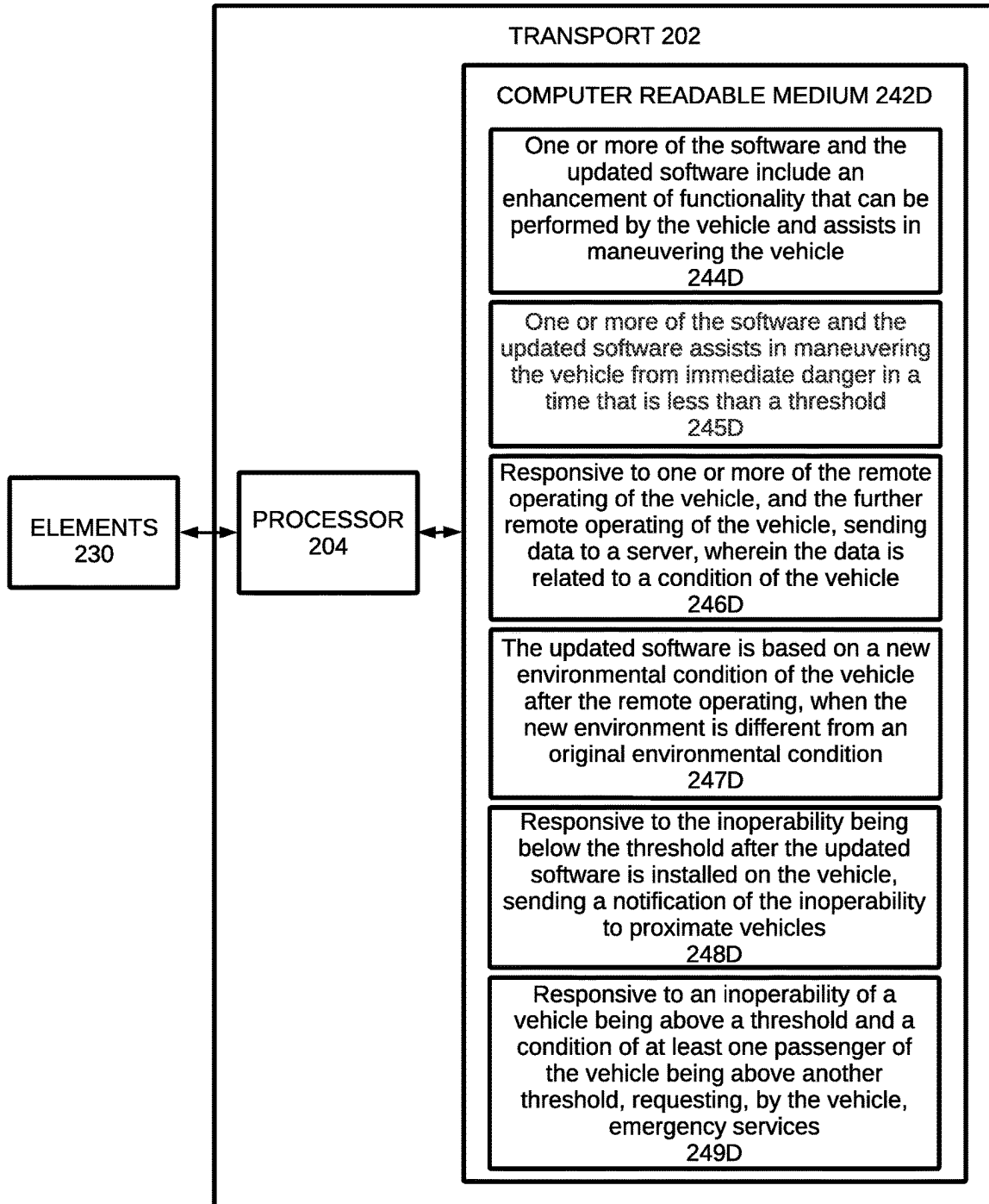
250

FIG. 2D

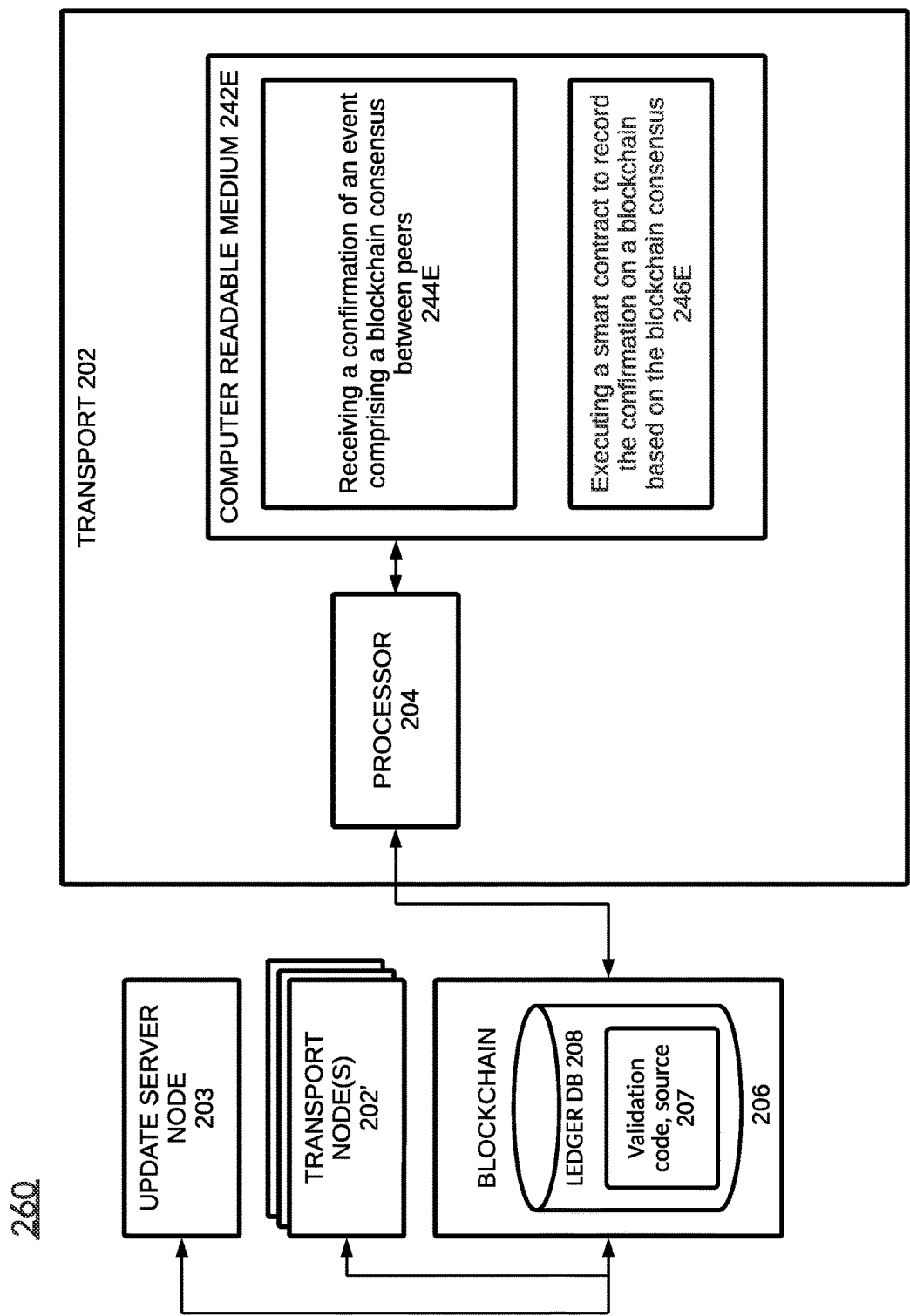


FIG. 2E

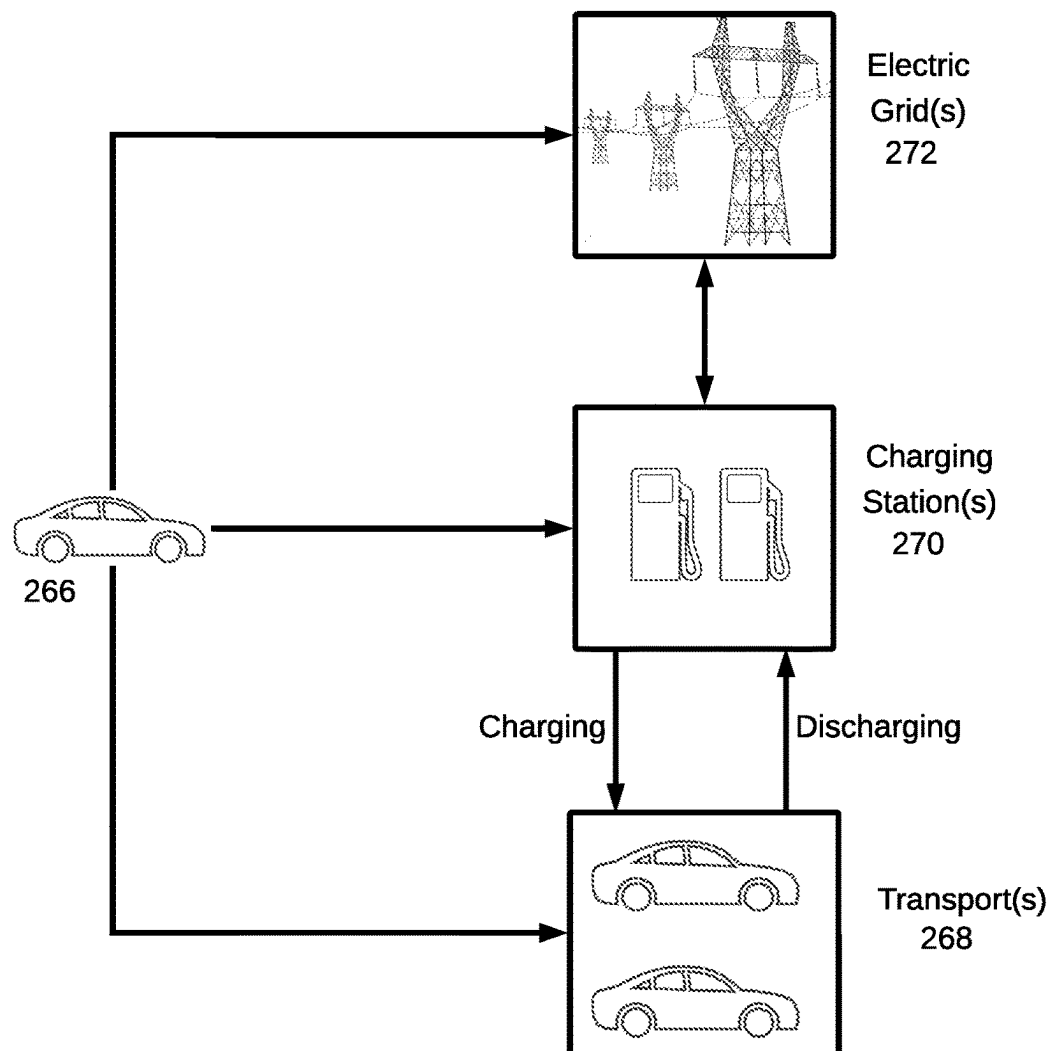
265

FIG. 2F

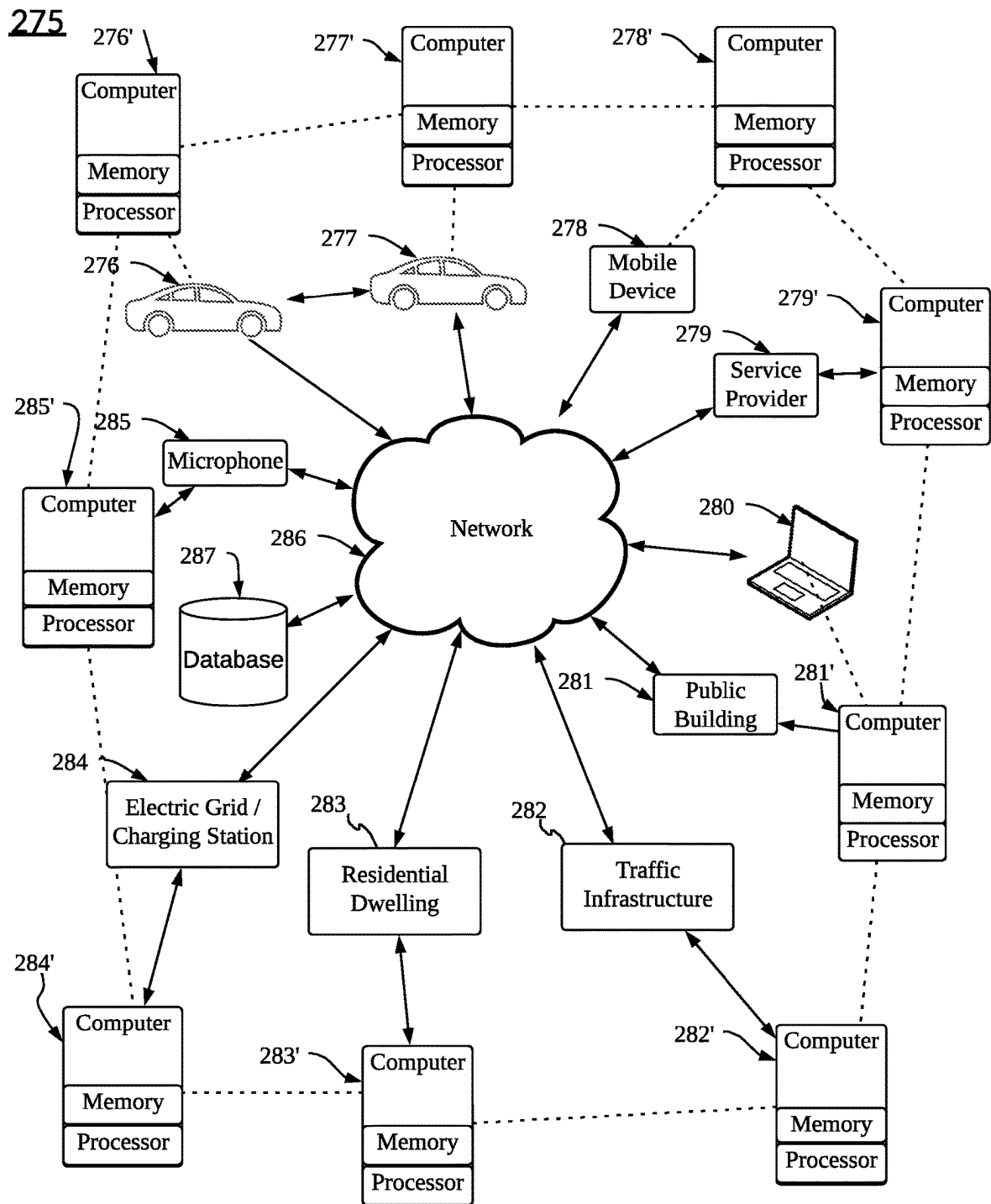


FIG. 2G

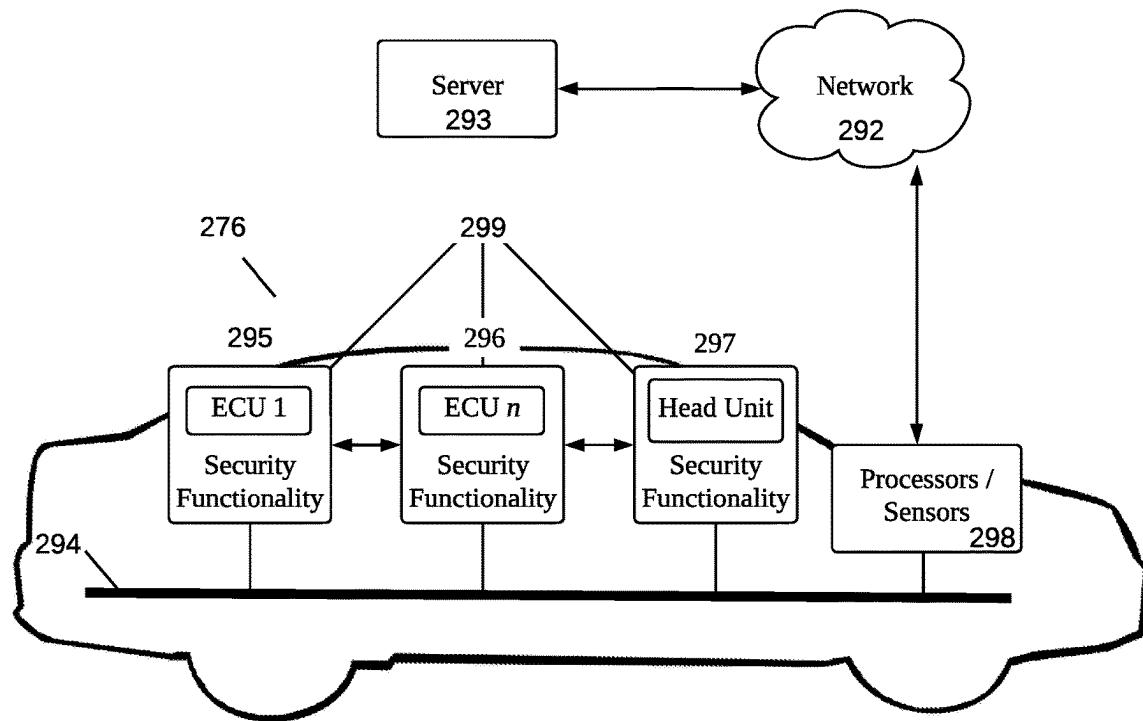
290

FIG. 2H

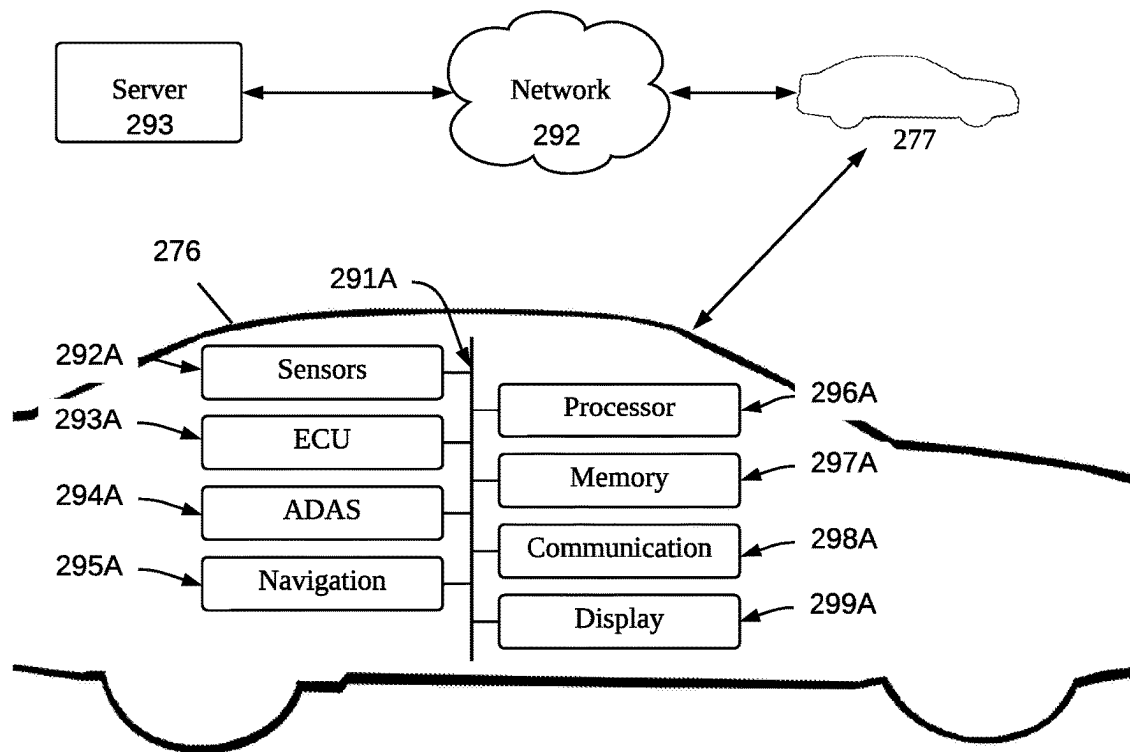
290A

FIG. 2I

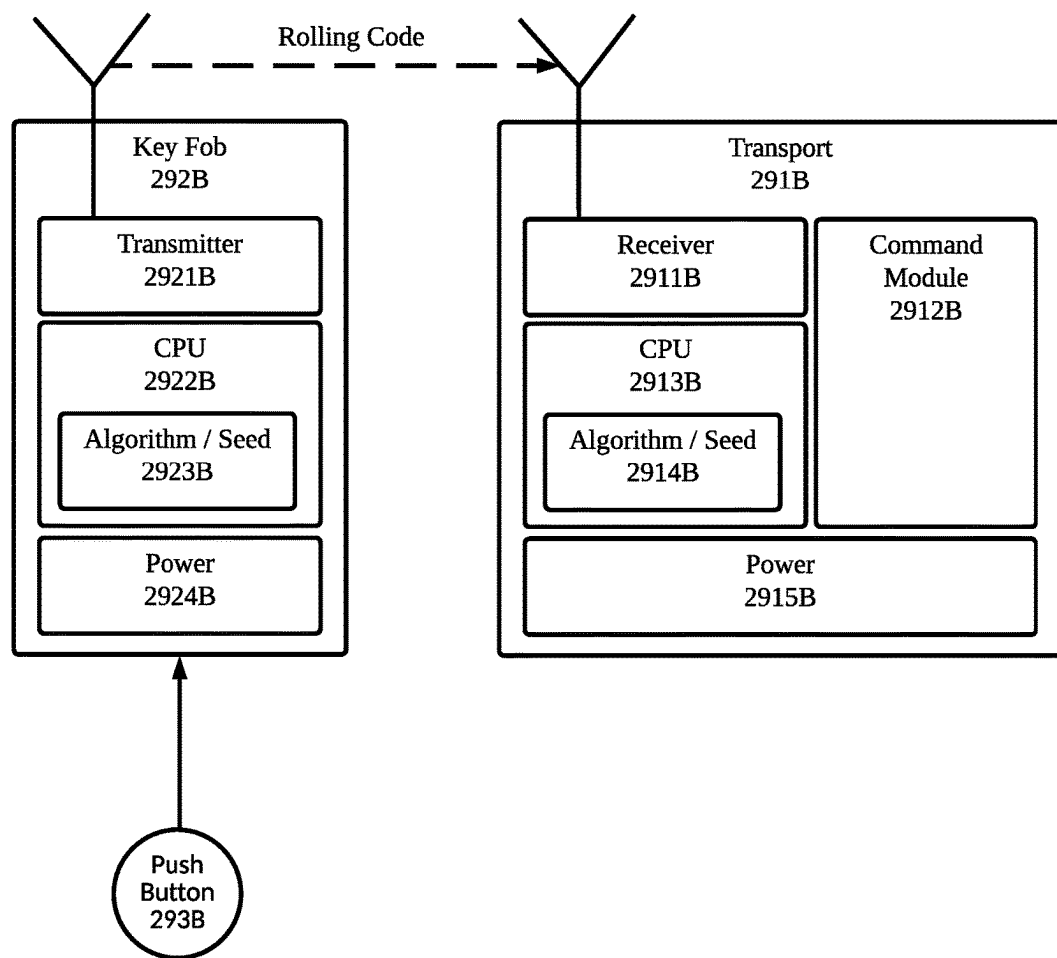
290B

FIG. 2J

290C

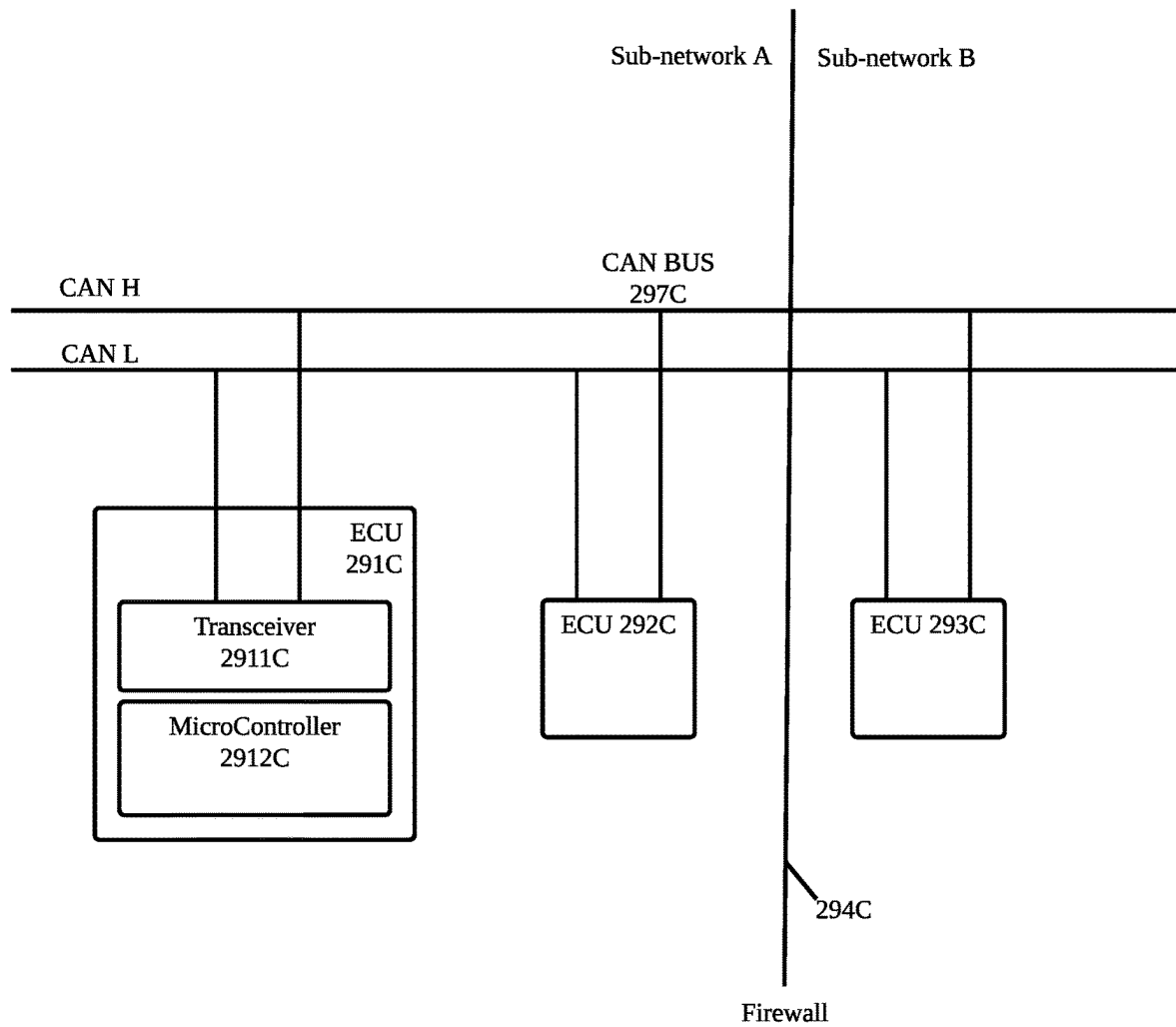


FIG. 2K

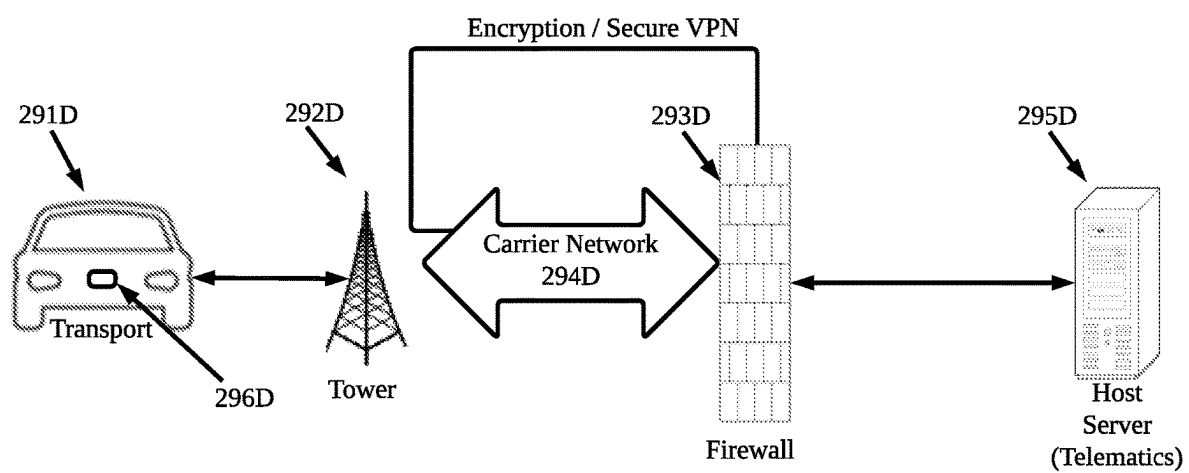
290D

FIG. 2L

290E

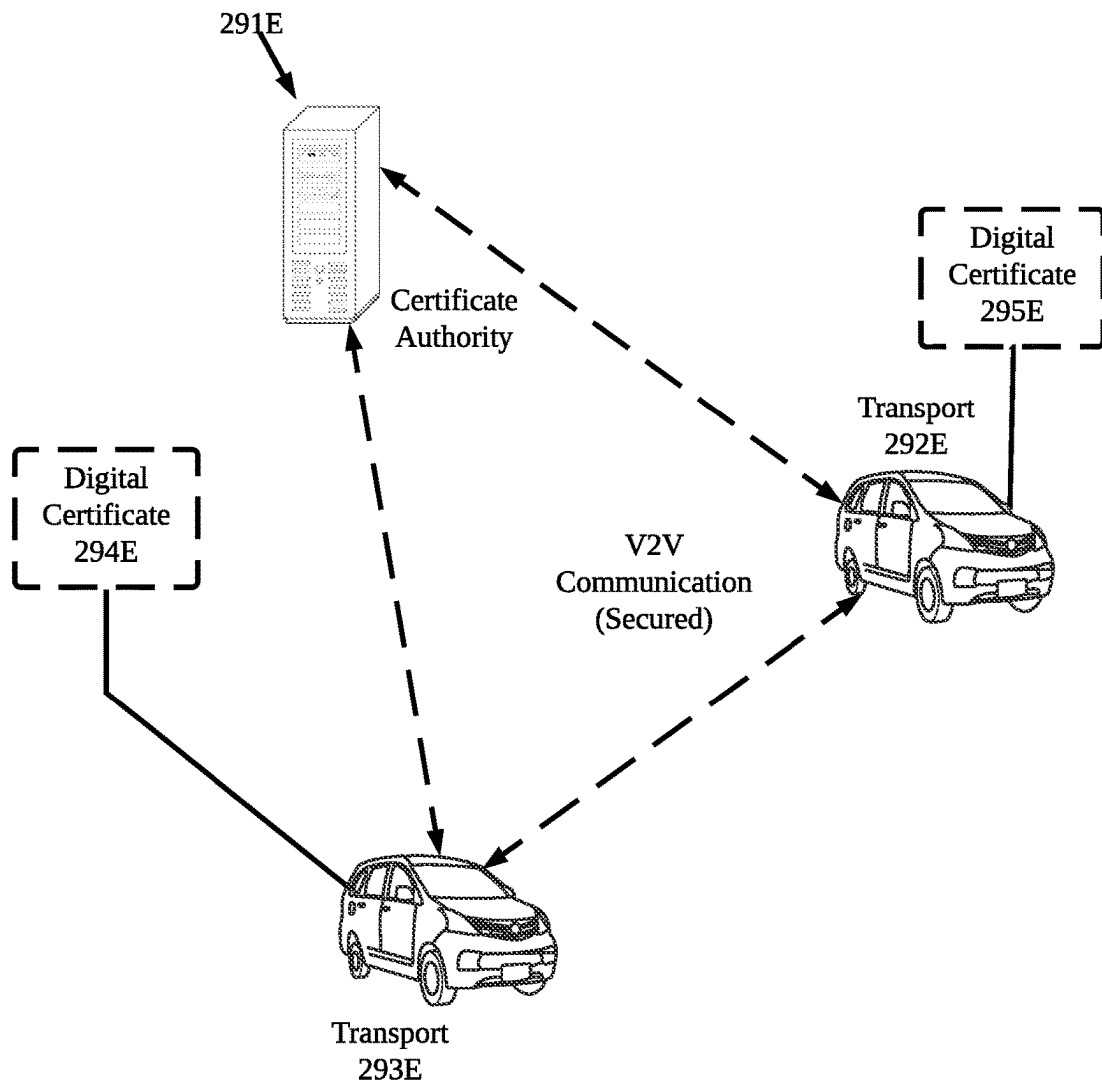


FIG. 2M

290E

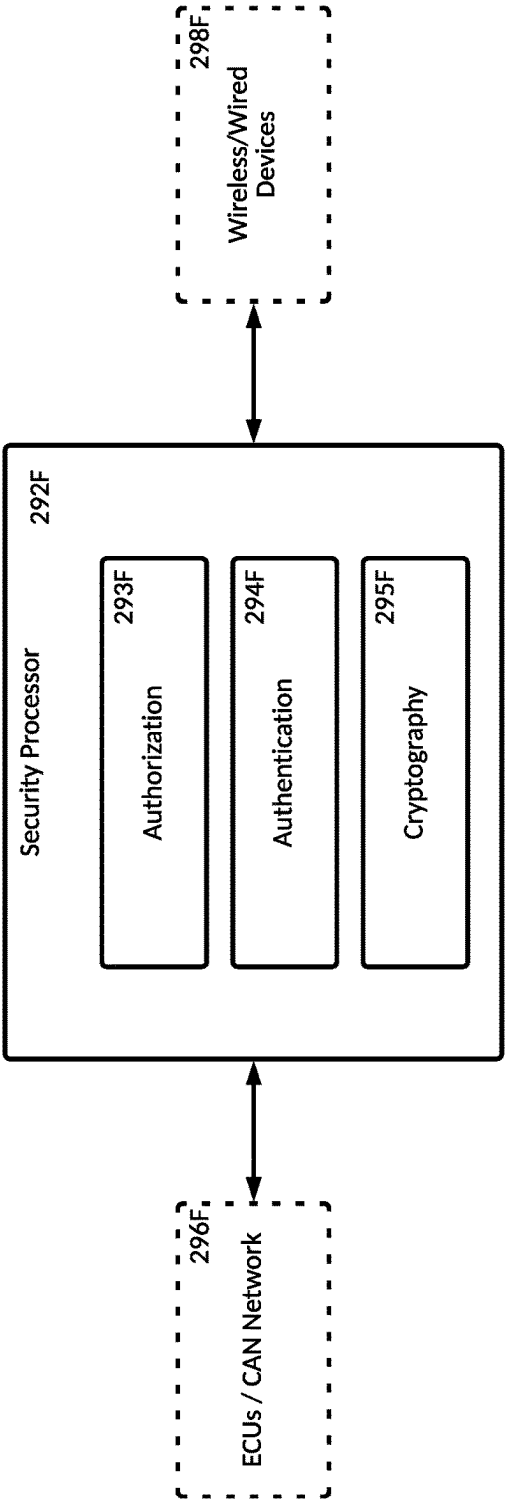


FIG. 2N

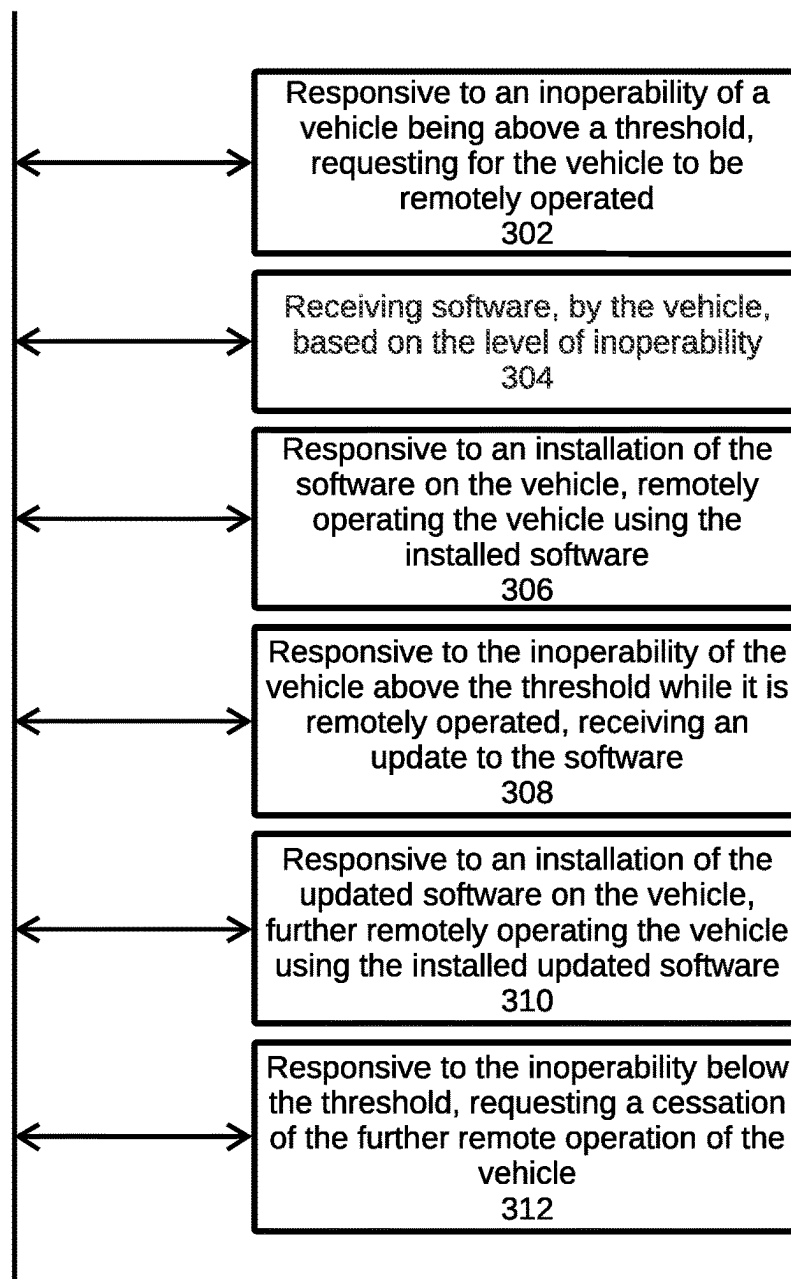
300

FIG. 3A

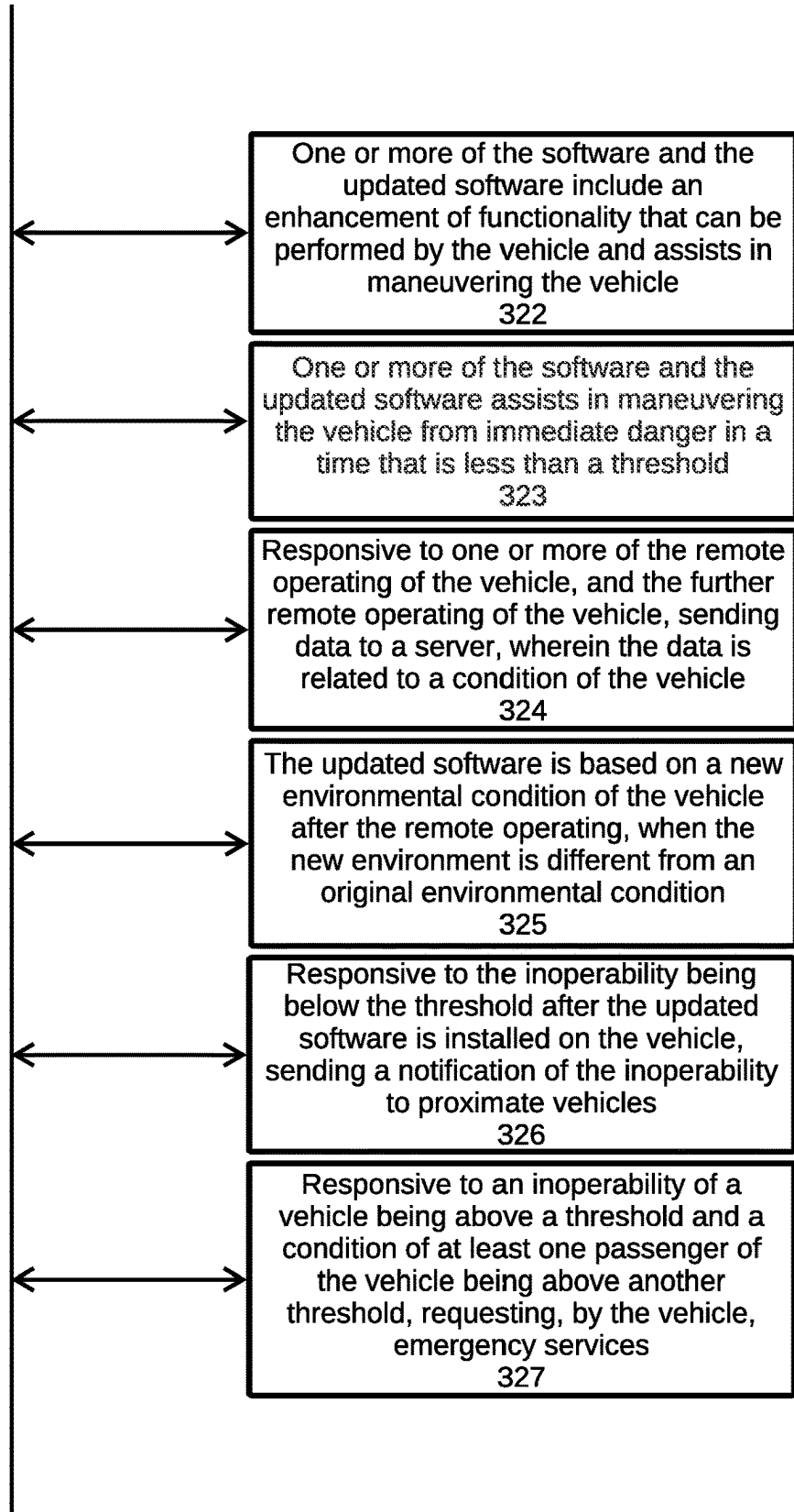
320

FIG. 3B

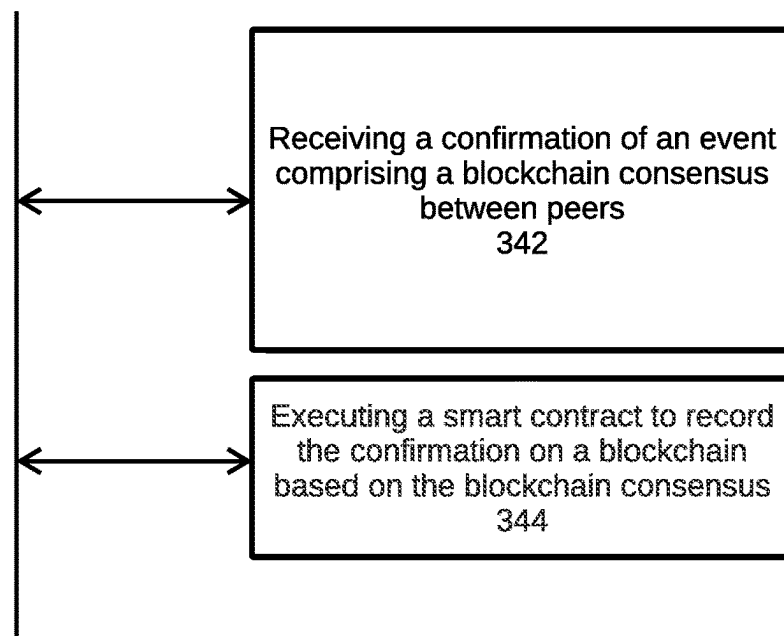
340

FIG. 3C

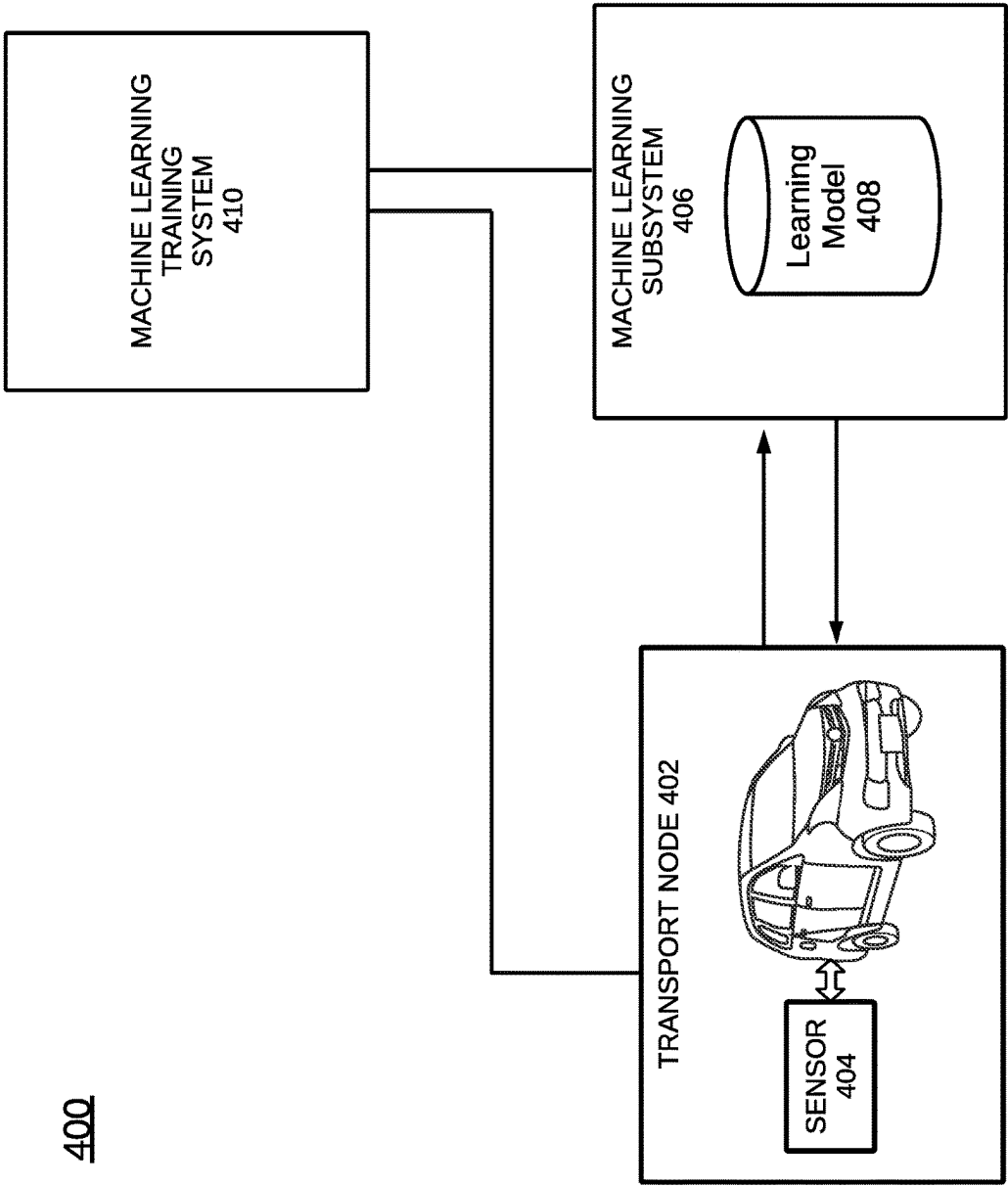


FIG. 4

500

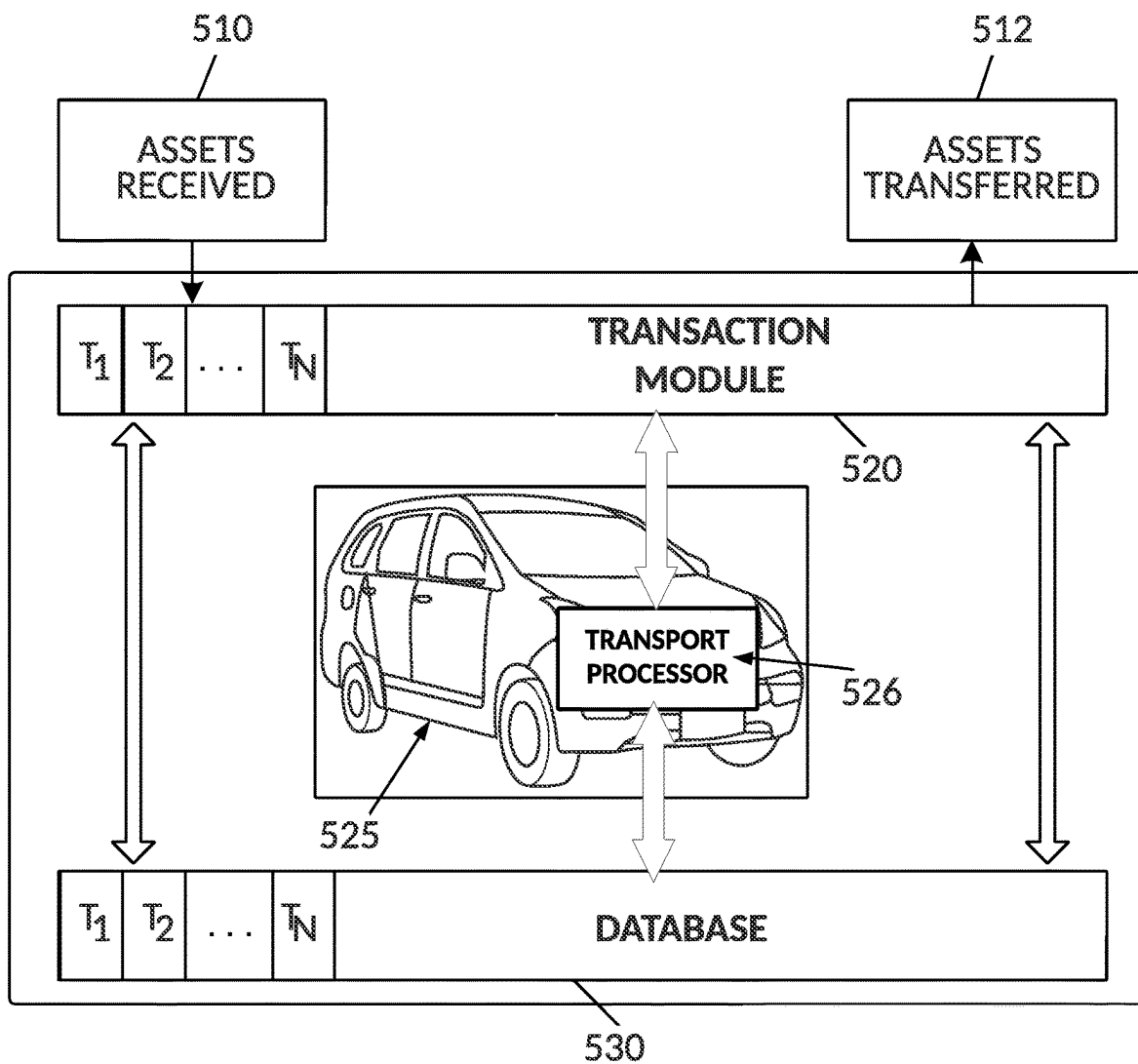


FIG. 5A

550

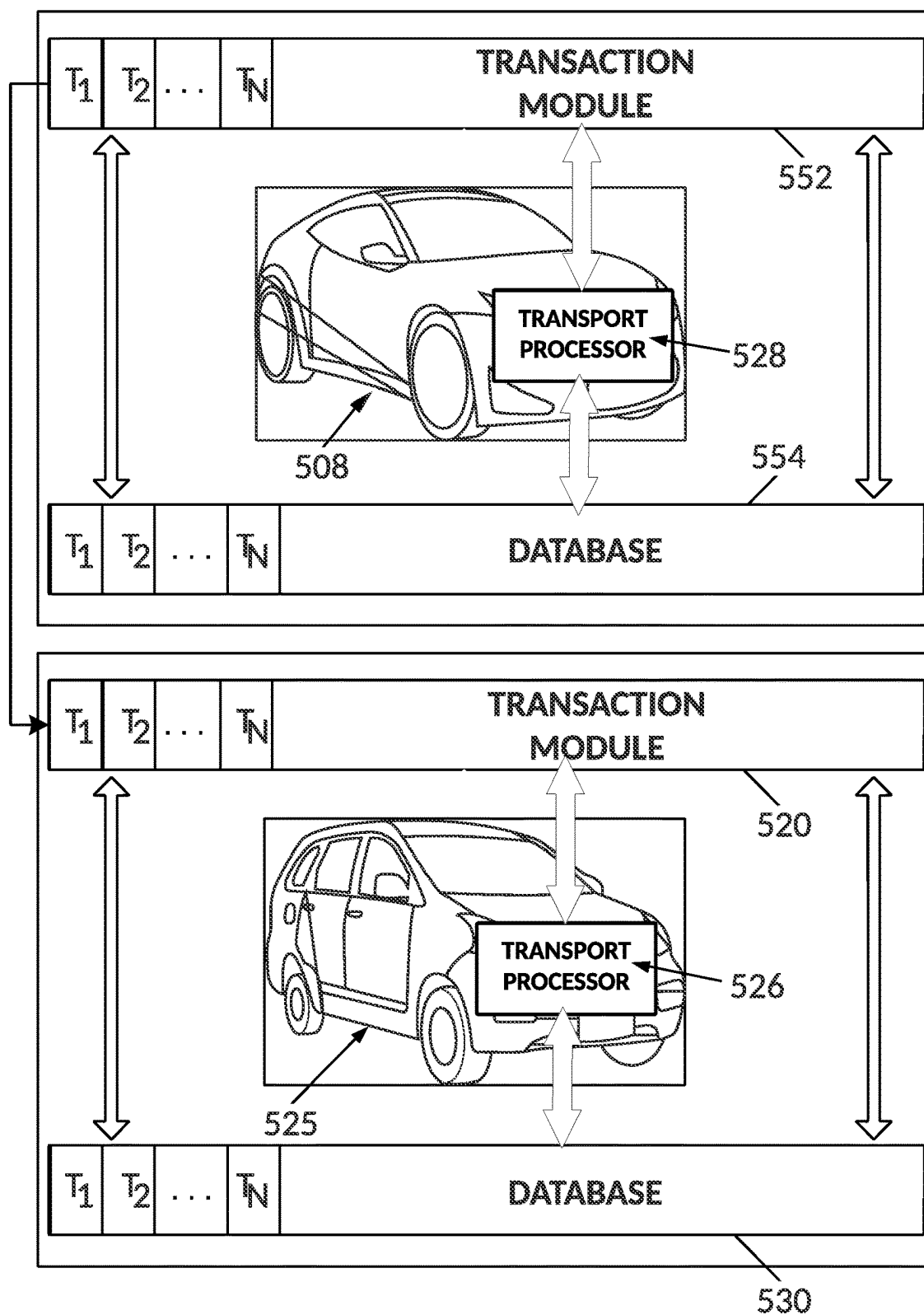


FIG. 5B

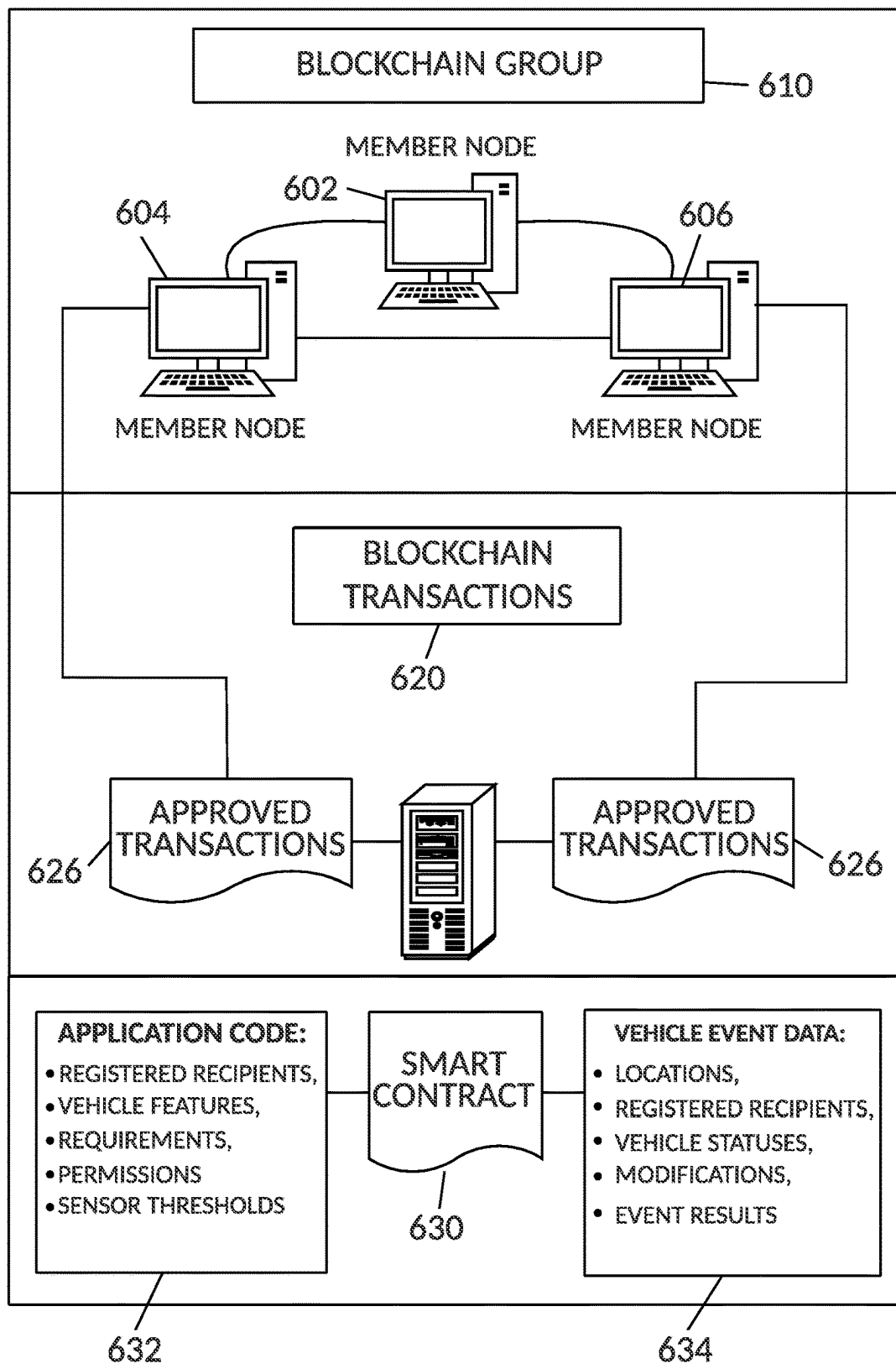
600

FIG. 6A

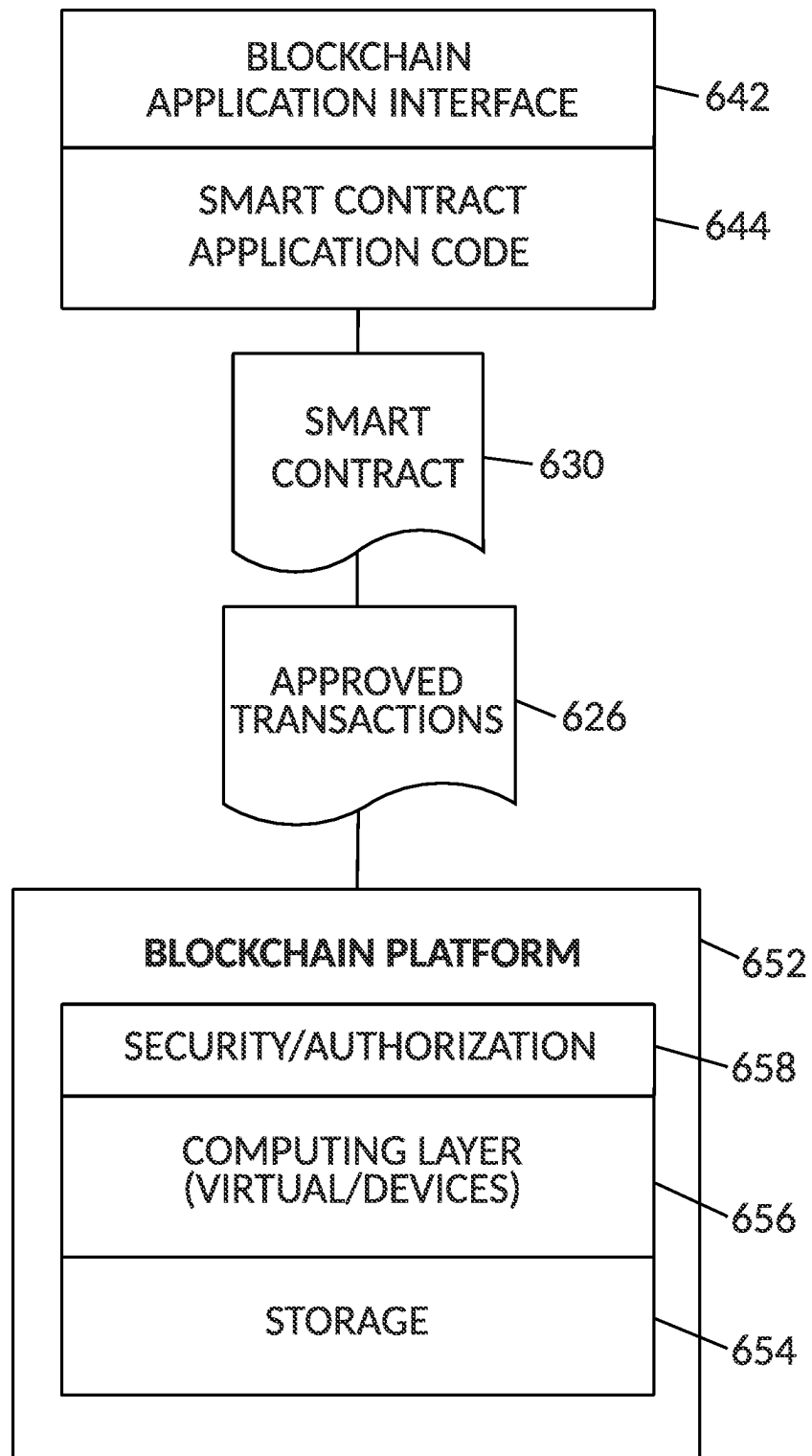
640

FIG. 6B

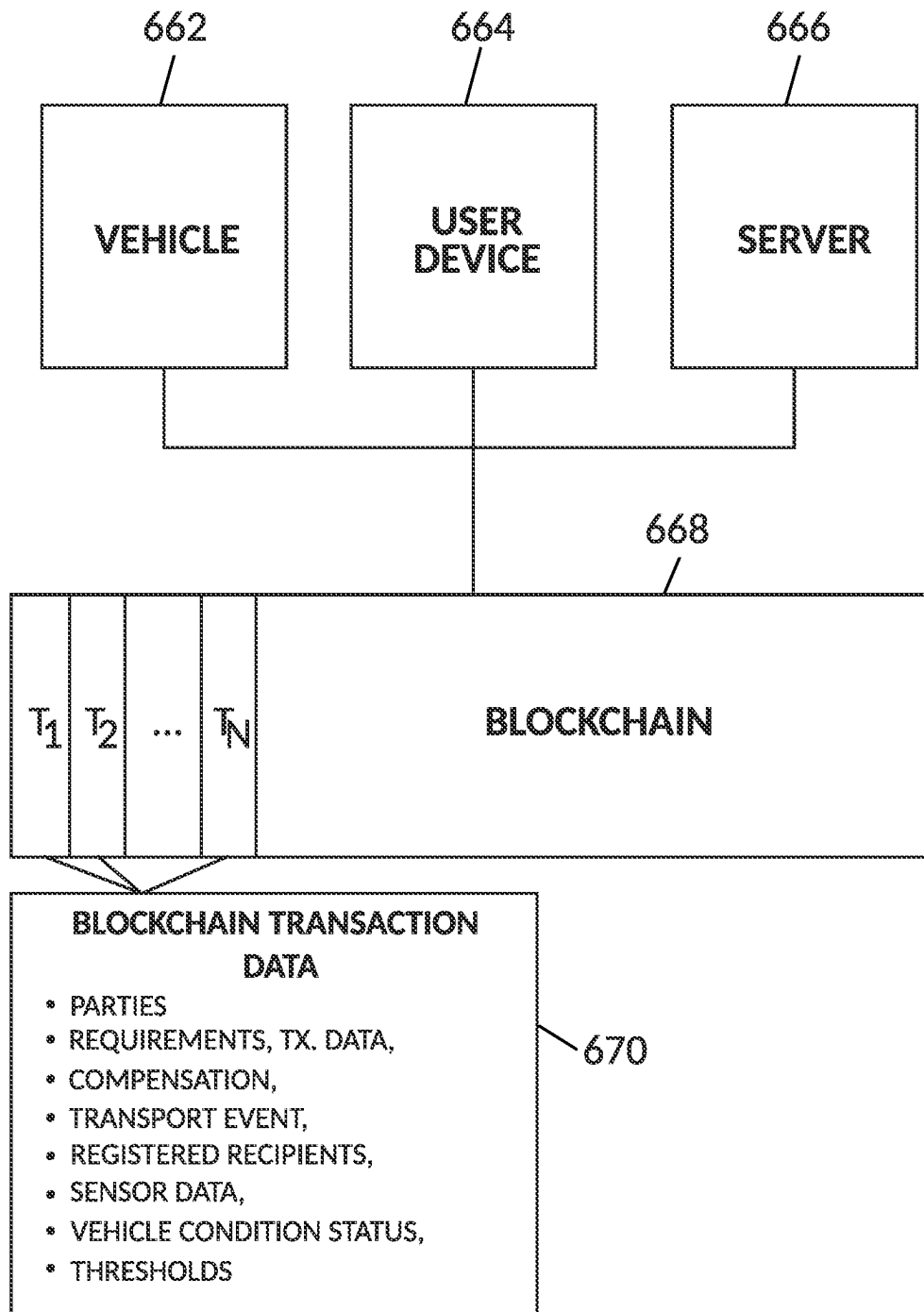
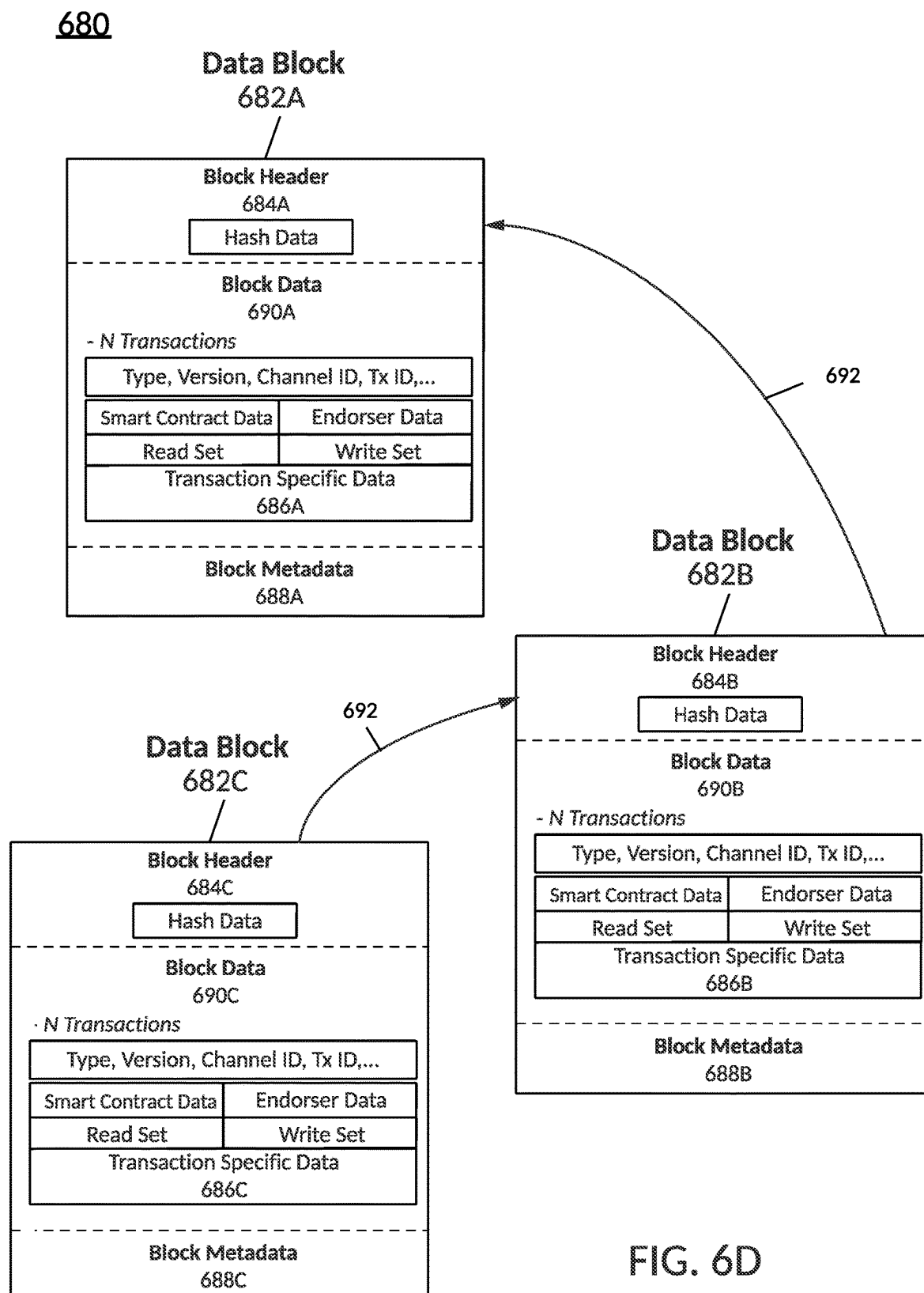
660

FIG. 6C



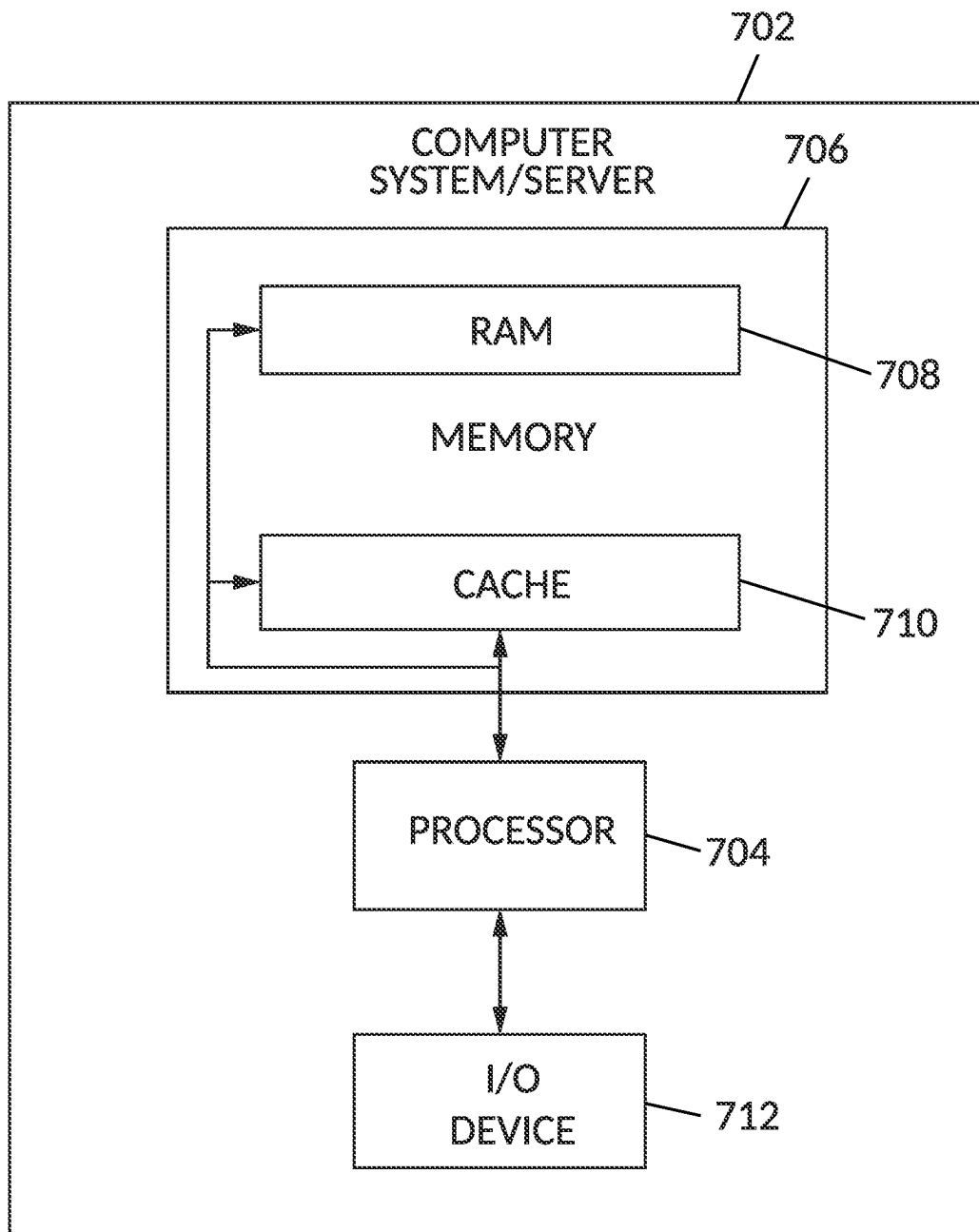
700

FIG. 7

1

DATA COMMUNICATION WITH REMOTE OPERATION OF A VEHICLE

BACKGROUND

Vehicles or transports, such as cars, motorcycles, trucks, planes, trains, etc., generally provide transportation needs to occupants and/or goods in a variety of ways. Functions related to transports may be identified and utilized by various computing devices, such as a smartphone or a computer located on and/or off the transport.

SUMMARY

One example embodiment provides a method that includes one or more of responsive to an inoperability of a vehicle above a threshold, requesting for the vehicle to be remotely operated, receiving software, by the vehicle, based on the level of inoperability, responsive to an installation of the software on the vehicle, remotely operating the vehicle using the installed software, responsive to the inoperability of the vehicle above the threshold while it is remotely operated, receiving an update to the software, responsive to an installation of the updated software on the vehicle, further remotely operating the vehicle using the installed updated software, and responsive to the inoperability below the threshold, requesting a cessation of the further remote operation of the vehicle.

Another example embodiment provides a system that includes a memory communicably coupled to a processor, wherein the processor performs one or more of responsive to an inoperability of a vehicle above a threshold, requests for the vehicle to be remotely operated, receives software, by the vehicle, based on the level of inoperability, responsive to an installation of the software on the vehicle, remotely operates the vehicle via the use of the installed software, responsive to the inoperability of the vehicle above the threshold, while it is remotely operated, receives an update to the software, responsive to an installation of the updated software on the vehicle, further remotely operates the vehicle using the installed updated software, and responsive to the inoperability below the threshold, requests a cessation of the further remote operation of the vehicle.

A further example embodiment provides a computer-readable storage medium comprising instructions, that when read by a processor, cause the processor to perform one or more of responsive to an inoperability of a vehicle above a threshold, requesting for the vehicle to be remotely operated, receiving software, by the vehicle, based on the level of inoperability, responsive to an installation of the software on the vehicle, remotely operating the vehicle using the installed software, responsive to the inoperability of the vehicle above the threshold while it is remotely operated, receiving an update to the software, responsive to an installation of the updated software on the vehicle, further remotely operating the vehicle using the installed updated software, and responsive to the inoperability below the threshold, requesting a cessation of the further remote operation of the vehicle.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A illustrates an example system diagram, according to example embodiments.

FIG. 1B illustrates an example flowchart, according to example embodiments.

2

FIG. 1C illustrates a further example flowchart, according to example embodiments.

FIG. 2A illustrates a transport network diagram, according to example embodiments.

FIG. 2B illustrates another transport network diagram, according to example embodiments.

FIG. 2C illustrates yet another transport network diagram, according to example embodiments.

FIG. 2D illustrates a further transport network diagram, according to example embodiments.

FIG. 2E illustrates yet a further transport network diagram, according to example embodiments.

FIG. 2F illustrates a diagram depicting electrification of one or more elements, according to example embodiments.

FIG. 2G illustrates a diagram depicting interconnections between different elements, according to example embodiments.

FIG. 2H illustrates a further diagram depicting interconnections between different elements, according to example embodiments.

FIG. 2I illustrates yet a further diagram depicting interconnections between elements, according to example embodiments.

FIG. 2J illustrates yet a further diagram depicting a keyless entry system, according to example embodiments.

FIG. 2K illustrates yet a further diagram depicting a CAN within a transport, according to example embodiments.

FIG. 2L illustrates yet a further diagram depicting an end-to-end communication channel, according to example embodiments.

FIG. 2M illustrates yet a further diagram depicting an example of transports performing secured V2V communications using security certificates, according to example embodiments.

FIG. 2N illustrates yet a further diagram depicting an example of a transport interacting with a security processor and a wireless device, according to example embodiments.

FIG. 3A illustrates a flow diagram, according to example embodiments.

FIG. 3B illustrates another flow diagram, according to example embodiments.

FIG. 3C illustrates yet another flow diagram, according to example embodiments.

FIG. 4 illustrates a machine learning transport network diagram, according to example embodiments.

FIG. 5A illustrates an example vehicle configuration for managing database transactions associated with a vehicle, according to example embodiments.

FIG. 5B illustrates another example vehicle configuration for managing database transactions conducted among various vehicles, according to example embodiments.

FIG. 6A illustrates a blockchain architecture configuration, according to example embodiments.

FIG. 6B illustrates another blockchain configuration, according to example embodiments.

FIG. 6C illustrates a blockchain configuration for storing blockchain transaction data, according to example embodiments.

FIG. 6D illustrates example data blocks, according to example embodiments.

FIG. 7 illustrates an example system that supports one or more of the example embodiments.

DETAILED DESCRIPTION

It will be readily understood that the instant components, as generally described and illustrated in the figures herein,

may be arranged and designed in a wide variety of different configurations. Thus, the following detailed description of the embodiments of at least one of a method, apparatus, computer-readable storage medium and system, as represented in the attached figures, is not intended to limit the scope of the application as claimed but is merely representative of selected embodiments. Multiple embodiments depicted herein are not intended to limit the scope of the solution. The computer-readable storage medium may be a non-transitory computer-readable medium or a non-transitory computer-readable storage medium.

Communications between the transport(s) and certain entities, such as remote servers, other transports and local computing devices (e.g., smartphones, personal computers, transport-embedded computers, etc.) may be sent and/or received and processed by one or more 'components' which may be hardware, firmware, software or a combination thereof. The components may be part of any of these entities or computing devices or certain other computing devices. In one example, consensus decisions related to blockchain transactions may be performed by one or more computing devices or components (which may be any element described and/or depicted herein) associated with the transport(s) and one or more of the components outside or at a remote location from the transport(s).

The instant features, structures, or characteristics described in this specification may be combined in any suitable manner in one or more embodiments. For example, the usage of the phrases "example embodiments," "some embodiments," or other similar language, throughout this specification refers to the fact that a particular feature, structure, or characteristic described in connection with the embodiment may be included in at least one example. Thus, appearances of the phrases "example embodiments", "in some embodiments", "in other embodiments," or other similar language, throughout this specification do not necessarily all refer to the same group of embodiments, and the described features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. In the diagrams, any connection between elements can permit one-way and/or two-way communication, even if the depicted connection is a one-way or two-way arrow. In the current solution, a vehicle or transport may include one or more of cars, trucks, walking area battery electric vehicle (BEV), e-Palette, fuel cell bus, motorcycles, scooters, bicycles, boats, recreational vehicles, planes, and any object that may be used to transport people and or goods from one location to another.

In addition, while the term "message" may have been used in the description of embodiments, other types of network data, such as, a packet, frame, datagram, etc. may also be used. Furthermore, while certain types of messages and signaling may be depicted in exemplary embodiments they are not limited to a certain type of message and signaling.

Example embodiments provide methods, systems, components, non-transitory computer-readable medium, devices, and/or networks, which provide at least one of a transport (also referred to as a vehicle or car herein), a data collection system, a data monitoring system, a verification system, an authorization system, and a vehicle data distribution system. The vehicle status condition data received in the form of communication messages, such as wireless data network communications and/or wired communication messages, may be processed to identify vehicle/transport status conditions and provide feedback on the condition and/or changes of a transport. In one example, a user profile may be

applied to a particular transport/vehicle to authorize a current vehicle event, service stops at service stations, to authorize subsequent vehicle rental services, and enable vehicle-to-vehicle communications.

Within the communication infrastructure, a decentralized database is a distributed storage system which includes multiple nodes that communicate with each other. A blockchain is an example of a decentralized database, which includes an append-only immutable data structure (i.e., a distributed ledger) capable of maintaining records between untrusted parties. The untrusted parties are referred to herein as peers, nodes, or peer nodes. Each peer maintains a copy of the database records, and no single peer can modify the database records without a consensus being reached among the distributed peers. For example, the peers may execute a consensus protocol to validate blockchain storage entries, group the storage entries into blocks, and build a hash chain via the blocks. This process forms the ledger by ordering the storage entries, as is necessary, for consistency. In public or permissionless blockchains, anyone can participate without a specific identity. Public blockchains can involve cryptocurrencies and use consensus-based on various protocols such as proof of work (PoW). Conversely, a permissioned blockchain database can secure interactions among a group of entities, which share a common goal, but which do not or cannot fully trust one another, such as businesses that exchange funds, goods, information, and the like. The instant solution can function in a permissioned and/or a permissionless blockchain setting.

Smart contracts are trusted distributed applications which leverage tamper-proof properties of the shared or distributed ledger (which may be in the form of a blockchain) and an underlying agreement between member nodes, which is referred to as an endorsement or endorsement policy. In general, blockchain entries are "endorsed" before being committed to the blockchain while entries, which are not endorsed are disregarded. A typical endorsement policy allows smart contract executable code to specify endorsers for an entry in the form of a set of peer nodes that are necessary for endorsement. When a client sends the entry to the peers specified in the endorsement policy, the entry is executed to validate the entry. After validation, the entries enter an ordering phase in which a consensus protocol produces an ordered sequence of endorsed entries grouped into blocks.

Nodes are the communication entities of the blockchain system. A "node" may perform a logical function in the sense that multiple nodes of different types can run on the same physical server. Nodes are grouped in trust domains and are associated with logical entities that control them in various ways. Nodes may include different types, such as a client or submitting-client node, which submits an entry-invocation to an endorser (e.g., peer), and broadcasts entry proposals to an ordering service (e.g., ordering node). Another type of node is a peer node, which can receive client submitted entries, commit the entries and maintain a state and a copy of the ledger of blockchain entries. Peers can also have the role of an endorser. An ordering-service-node or orderer is a node running the communication service for all nodes and which implements a delivery guarantee, such as a broadcast to each of the peer nodes in the system when committing entries and modifying a world state of the blockchain. The world state can constitute the initial blockchain entry, which normally includes control and setup information.

A ledger is a sequenced, tamper-resistant record of all state transitions of a blockchain. State transitions may result

from smart contract executable code invocations (i.e., entries) submitted by participating parties (e.g., client nodes, ordering nodes, endorser nodes, peer nodes, etc.). An entry may result in a set of asset key-value pairs being committed to the ledger as one or more operands, such as creates, updates, deletes, and the like. The ledger includes a blockchain (also referred to as a chain), which stores an immutable, sequenced record in blocks. The ledger also includes a state database, which maintains a current state of the blockchain. There is typically one ledger per channel. Each peer node maintains a copy of the ledger for each channel of which they are a member.

A chain is an entry log structured as hash-linked blocks, and each block contains a sequence of N entries where N is equal to or greater than one. The block header includes a hash of the blocks' entries, as well as a hash of the prior block's header. In this way, all entries on the ledger may be sequenced and cryptographically linked together. Accordingly, it is not possible to tamper with the ledger data without breaking the hash links. A hash of a most recently added blockchain block represents every entry on the chain that has come before it, making it possible to ensure that all peer nodes are in a consistent and trusted state. The chain may be stored on a peer node file system (i.e., local, attached storage, cloud, etc.), efficiently supporting the append-only nature of the blockchain workload.

The current state of the immutable ledger represents the latest values for all keys that are included in the chain entry log. Since the current state represents the latest key values known to a channel, it is sometimes referred to as a world state. Smart contract executable code invocations execute entries against the current state data of the ledger. To make these smart contract executable code interactions efficient, the latest values of the keys may be stored in a state database. The state database may be simply an indexed view into the chain's entry log and can therefore be regenerated from the chain at any time. The state database may automatically be recovered (or generated if needed) upon peer node startup and before entries are accepted.

A blockchain is different from a traditional database in that the blockchain is not a central storage but rather a decentralized, immutable, and secure storage, where nodes must share in changes to records in the storage. Some properties that are inherent in blockchain and which help implement the blockchain include, but are not limited to, an immutable ledger, smart contracts, security, privacy, decentralization, consensus, endorsement, accessibility, and the like.

Example embodiments provide a service to a particular vehicle and/or a user profile that is applied to the vehicle. For example, a user may be the owner of a vehicle or the operator of a vehicle owned by another party. The vehicle may require service at certain intervals, and the service needs may require authorization before permitting the services to be received. Also, service centers may offer services to vehicles in a nearby area based on the vehicle's current route plan and a relative level of service requirements (e.g., immediate, severe, intermediate, minor, etc.). The vehicle needs may be monitored via one or more vehicle and/or road sensors or cameras, which report sensed data to a central controller computer device in and/or apart from the vehicle. This data is forwarded to a management server for review and action. A sensor may be located on one or more of the interior of the transport, the exterior of the transport, on a fixed object apart from the transport, and on another transport proximate the transport. The sensor may also be associated with the transport's speed, the transport's braking, the

transport's acceleration, fuel levels, service needs, the gear-shifting of the transport, the transport's steering, and the like. A sensor, as described herein, may also be a device, such as a wireless device in and/or proximate to the transport. Also, sensor information may be used to identify whether the vehicle is operating safely and whether an occupant has engaged in any unexpected vehicle conditions, such as during a vehicle access and/or utilization period. Vehicle information collected before, during and/or after a vehicle's operation may be identified and stored in a transaction on a shared/distributed ledger, which may be generated and committed to the immutable ledger as determined by a permission granting consortium, and thus in a "decentralized" manner, such as via a blockchain membership group.

Each interested party (i.e., owner, user, company, agency, etc.) may want to limit the exposure of private information, and therefore the blockchain and its immutability can be used to manage permissions for each particular user vehicle profile. A smart contract may be used to provide compensation, quantify a user profile score/rating/review, apply vehicle event permissions, determine when service is needed, identify a collision and/or degradation event, identify a safety concern event, identify parties to the event and provide distribution to registered entities seeking access to such vehicle event data. Also, the results may be identified, and the necessary information can be shared among the registered companies and/or individuals based on a consensus approach associated with the blockchain. Such an approach could not be implemented on a traditional centralized database.

Various driving systems of the instant solution can utilize software, an array of sensors as well as machine learning functionality, light detection and ranging (Lidar) projectors, radar, ultrasonic sensors, etc. to create a map of terrain and road that a transport can use for navigation and other purposes. In some embodiments, GPS, maps, cameras, sensors and the like can also be used in autonomous vehicles in place of Lidar.

The instant solution includes, in certain embodiments, authorizing a vehicle for service via an automated and quick authentication scheme. For example, driving up to a charging station or fuel pump may be performed by a vehicle operator or an autonomous transport and the authorization to receive charge or fuel may be performed without any delays provided the authorization is received by the service and/or charging station. A vehicle may provide a communication signal that provides an identification of a vehicle that has a currently active profile linked to an account that is authorized to accept a service, which can be later rectified by compensation. Additional measures may be used to provide further authentication, such as another identifier may be sent from the user's device wirelessly to the service center to replace or supplement the first authorization effort between the transport and the service center with an additional authorization effort.

Data shared and received may be stored in a database, which maintains data in one single database (e.g., database server) and generally at one particular location. This location is often a central computer, for example, a desktop central processing unit (CPU), a server CPU, or a mainframe computer. Information stored on a centralized database is typically accessible from multiple different points. A centralized database is easy to manage, maintain, and control, especially for purposes of security because of its single location. Within a centralized database, data redundancy is minimized as a single storing place of all data also implies

that a given set of data only has one primary record. A blockchain may be used for storing transport-related data and transactions.

Any of the actions described herein may be performed by one or more processors (such as a microprocessor, a sensor, an Electronic Control Unit (ECU), a head unit, and the like), with or without memory, which may be located on-board the transport and/or off-board the transport (such as a server, computer, mobile/wireless device, etc.). The one or more processors may communicate with other memory and/or other processors on-board or off-board other transports to utilize data being sent by and/or to the transport. The one or more processors and the other processors can send data, receive data, and utilize this data to perform one or more of the actions described or depicted herein.

FIG. 1A illustrates a system diagram **100** in one set of embodiments. A vehicle **102** is shown, containing a processor **104** and sensors **106**. A vehicle processor **104** may include a central vehicle processor, such as an Electronic Control Module (ECM). It may be communicably coupled to one or more other vehicle processors, including one or more processors associated with performing methods of the present application. Vehicle processors **104** may also be communicably coupled to the processors of other vehicles, servers, and the like. In one embodiment, a vehicle processor **104** may transmit and receive messages through wireless communications interfaces to and from the other processors, such as processors in a server **108**, another vehicle, and/or occupant devices. In another embodiment, a vehicle processor **104** may transmit and receive messages through wired communications interfaces, e.g., through a wired communication interface associated with a charging station connected to the vehicle, through a wired charging interface such as a universal serial bus USB, a connection for an occupant device, and the like, to and from another vehicle and/or other processors. Vehicle processors **104** may have one or more associated memory devices for storing applications and data and can be interconnected by various wired or wireless communications pathways, such as a Controller Area Network (CAN) bus or different wireless technologies known in the art.

Vehicle **102** may have sensors **106** that capture data from the vehicle's interior or exterior, such as the vehicle cabin, and exterior such as the area proximate the outside of vehicle **102**. Sensors **106** may include cameras, radar, microphones, and the like, that capture data and may send the captured data to the vehicle processor **104** for analysis and/or processing. Sensors **106** may also communicate with other processors, such as processors associated with various systems of vehicle **102**. For example, the braking system may contain sensors wherein the data from the sensors communicate with a processor associated with the braking system, or the sensors directly communicate with the vehicle processor, wherein communication may occur over a bus, such as the CAN bus. This is a message-based protocol designed specifically for automotive applications that allow micro-controllers and devices to communicate with each other's applications without a host computer.

The system may also include one or more servers **108**. Servers **108** may include one or more computers communicably coupled to the vehicle processor **104**, the sensors **106**, and/or any other processor associated with vehicle **102**, which may route communication through a network **112**, which may be referred to as the Cloud or Internet. Servers **108** may include one or more processors and memory devices for storing applications and data. In one embodiment, servers **108** may be associated with a vehicle manu-

facturer, including a repair facility, a sales facility, an entertainment establishment, and the like. In one embodiment, servers **108** may be in a network **112**, a cloud, and/or in or connected to a vehicle charging station.

In some embodiments, the instant solution fully or partially executes in a memory of a processor **104** of a computer associated with a vehicle **102**, in a memory of a server **108**, and/or in a memory of one or more other processors associated with devices and/or entities mentioned herein. One or more of the server **108** and the processor **104** may be communicably coupled to a network **112**. In some embodiments, the instant solution executes fully or partially on any processor or server located on any element in the system **100**.

In one example of the current solution, the processor may determine a level of inoperability of a vehicle **102** after an event. The event may be an occurrence where there is damage to vehicle **102**. Sensors **106** on the vehicle may detect the damage and may include impact detection devices, microphones, etc., and communicate with the processor **104**. A damage threshold may be established and stored in the memory associated with the processor **104**. This threshold may be used by the instant solution to determine a level of inoperability and/or a severity of damage to the vehicle **102** and may include several sensors **106** that fail to communicate with the processor **104** or receive incoherent or incorrect data from the sensors **106**. In an alternate embodiment, the threshold may be stored remotely in vehicle **102**, such as in server **108**. A message, such as a Remote Request+Vehicle Data Message, may be sent from processor **104** to server **108** when the determined damage to vehicle **102** exceeds the threshold. The message may include data collected by the processor **104**, data from the sensors **106**, and/or a summary of the inoperability and/or damage. Server **108**, associated with vehicle **102**, may receive the remote request message from vehicle **102**. It may also determine the appropriate software needed to operate the vehicle remotely, modify it if necessary, and use it to control it remotely. In another embodiment, the software is sent to vehicle **102**, based on the inoperability and/or damage, for example. The software may include context-based dynamical model data software that assists in remotely operating the vehicle. It could be a kinematic bicycle model, a bicycle model with tires, a longitudinal model, or a Carsim multiple degrees of freedom model. The software may be updated and/or modified as necessary based on data, which may be continually received from vehicle **102**, and may be sent to vehicle **102** by server **108** for installation. Vehicle **102** may send a notification of the installation of the software to server **108**. A remote operator **110** may remotely operate the vehicle and may utilize the software and/or the updated software sent by server **108**. This system involves vehicle **102**, equipped with sensors **106** and a processor **104** that communicates with a server **108** to evaluate vehicle damage and facilitate remote operation **110** when needed. Server **108** may determine and send the necessary software for the remote operation **110** of the vehicle. The remote operator **110** may be a separate entity from server **108** or may be one and the same.

Now referring to FIG. 1B, a flowchart **120** is depicted, according to an example embodiment. Vehicle **102** is involved in a collision or other event, rendering vehicle **102** inoperable. The event may cause an indentation on the outside portion of the vehicle, like the frontend bumper, an exterior panel, etc. The sensors **106** mounted on the vehicle may detect the impact. These sensors may include accelerometers measuring the amount of movement during the impact,

microphones recording the sound of the impact, and the like. The sensors **106** may communicate **122** the collected data to processor **104** through a vehicle network protocol and may occur over the CAN bus. The processor **104** on the vehicle may analyze the received data to determine the level of inoperability and/or damage **124**. This analysis might include comparing the sensor readings against a preset threshold stored in memory associated with the vehicle processor **104** and/or a server **108**. This threshold might represent a level of impact or deformation that is deemed to be significant, for example.

The vehicle's processor **104** may monitor the communication from the sensors **106**, such as over the CAN bus. If a sensor stops communicating or sends erratic data, processor **104** may interpret this as sensor **106** being damaged or disconnected. The processor **104** may update its assessment of the vehicle's damage level to account for any potentially damaged or disconnected sensors. Based on the determined level of damage **124** and the threshold comparison **126**, processor **104** may send a message (such as a Remote Request+Vehicle Data message **128**) to an external server **108** to indicate the inoperability and/or the severity of the damage and request a remote operation of vehicle **102**. The external server **108** may be associated with the vehicle manufacturer or a dedicated vehicle monitoring service. In another embodiment, other messages may be sent to other processors in vehicle **102** that may trigger several possible responses, including warning lights on the dashboard, messages to the driver, or even automatic emergency calls.

Server **108** may identify software **130** needed to enable the remote operation of vehicle **102** safely. This software might include a dynamical model that considers the state of the vehicle, including its damage level, and may help predict its behavior under various conditions. The specific type of model used (kinematic bicycle model, bicycle model with tires, longitudinal model, Carsim multiple degrees of freedom model, etc.) may depend on the specific circumstances and the vehicle's design, and may be determined by the received data **128**.

Server **108** may modify this software based on the received damage data to account for the vehicle's current state. The server may send a message, such as a Request Data message **132**, received at processor **104** on vehicle **102**. The processor **104**, responsive to the received Request Data message **132**, may obtain current data **133** from vehicle **102** and send the data back to server **108** via a message, such as Vehicle Data message **134**. For example, data from sensors **106** may be obtained. Additionally, other data may be obtained from other processors on vehicle **102**, such as other ECUs that may contain data related to different systems in the vehicle. For example, systems that control airbags, braking, steering, acceleration, etc. The data may also include the PSI of the tires, the vehicle's **102** weight, acceleration and/or braking data, and the like. The processor **104** may also determine a dynamics model that vehicle **102** can use, considering the current state of vehicle **102**. For example, in an icy road scenario, a dynamic bicycle model may be determined to be the best model to utilize. This model may determine a coefficient of friction corresponding to the icy conditions of one or more of the vehicle's wheels. The returned vehicle data in Vehicle Data message **134** may include at least data obtained by sensors and/or other processors in vehicle **102**, metadata, parameters, etc., and the recommended vehicle dynamics model collected by the vehicle processor **104**.

Server **108** may determine **136** if the originally determined vehicle dynamics model **130** differs from the vehicle

dynamics model returned in Vehicle Data message **134** and/or determine that a new vehicle dynamics model is needed based on the received data obtained by the sensors and/or other processors in vehicle **102**, metadata, and/or parameters, etc. If server **108** determines that a new model is needed, a new model is created **138** and may be implemented in a Graphical User Interface (GUI) **140** to be utilized by a remote-control operator, for example. The original or new vehicle dynamics model may assist the remote operator **140** in controlling vehicle **102** based on the situation that vehicle **102** is currently in.

In one embodiment, server **108** may continue **144** to request current data of vehicle **102** by the sending of a message, such as Request Data message **132**, wherein vehicle processor **104** obtains current data as further disclosed herein and returns the collected data in a message sent back to server **108**. This may be performed by server **108** on an interval, such as every 5 minutes and/or when necessary, based on the maneuvering of vehicle **102**, by the remote operator.

Referring to FIG. 1C, another flowchart **150** is depicted, according to another example embodiment. Vehicle **102** is involved in an event, such as a collision or a scenario where it is difficult or impossible to maneuver out of. The sensors **106** mounted on the vehicle may detect the impact. These could include accelerometers measuring the amount of movement during the impact, microphones recording the sound of the impact, and more. The sensors **106** may communicate **152** the collected data to a vehicle processor **104** through a vehicle network protocol via the CAN bus, for example. The processor **104** may analyze the received sensor data **152** to determine the inoperability of vehicle **102**, which may include a determination of the level of damage **154**. This analysis might include comparing the sensor readings against a preset threshold stored in memory associated with the vehicle processor **104** and/or a server **108**. This threshold might represent an inability of vehicle **102** to be operated and/or a level of impact or deformation that is deemed to be significant, for example. The inoperability of vehicle **102** may be associated with sensor data indicating a lack of traction control, an inability of a steering capability, a lack of movement, etc.

The vehicle's processor **104** may monitor the communication from the sensors **106** over the CAN bus. If a sensor stops communicating or sends erratic data, processor **104** may interpret this as sensor **106** being damaged or disconnected. The processor may update its assessment of the vehicle's damage level to account for any potentially damaged or disconnected sensors. Based on the determined level of damage **154** and the threshold comparison **156**, processor **104** may send a message (such as a Remote Request+Vehicle Data message **158**) to an external server **108**, wherein the data may include sensor **106** data of vehicle **102**, such as media from the sensors **106**, data related to various systems of vehicle **102** which may include tire traction, steering data, braking, and acceleration data, the severity of damage to vehicle **102**, and the like. The message may request a remote operation of vehicle **102**. The message may include data collected by processor **104**, such as data from sensors **106**, and/or other processors on vehicle **102**, such as ECUs. The external server **108** may be associated with the vehicle manufacturer or a dedicated vehicle monitoring service.

The server **108** may determine software **160** based on the received data in the remote request message **158**. The received data may also contain information about the make, model, year, included, and optional accessories existing on

11

vehicle 102, for example. The software may include one or more updates to the vehicle, such as additional traction control, modifications that may be applied to different systems of vehicle 102, such as braking and steering modifications, and may be based on the current situation of vehicle 102. For example, vehicle 102 is finding it difficult to maneuver on an icy road. The software may provide intelligence to the traction control system that provides additional traction to one or more wheels, allowing vehicle 102 to maneuver. The software is sent to vehicle 102 via a message, such as an Update Vehicle message 162. Processor 104 receives the message and updates 164 vehicle 102, which may include an update of software that resides in memory associated with processor 104 and/or other processors on vehicle 102, such as other ECUs, sensors, and the like. Vehicle 102 may be remotely operated 166 by a remote operator associated with server 108, which may be performed by a processor associated with server 108 and/or a remote operator. Vehicle 102 may begin maneuvering 170 by the remote operation, wherein communication 168 may occur between processor 104 and server 108.

Vehicle 102 may continue to send vehicle data 172 to server 108 to update the current condition of vehicle 102. A message, such as an Updated Vehicle Data message 172, is sent to server 108. This may include data pertaining to received sensor 106 data and/or data from other processors in vehicle 102, such as ECUs, etc. The Updated Vehicle Data message 172 may be sent on an interval, such as every minute, or processor 104 may send the message upon a determination that received sensor 106 data differs from previously recorded sensor 106 data. For example, sensors 106 and/or an ECU providing traction-related data from all wheels on vehicle 102 may collect and send data to processor 104, wherein processor 104 may determine that a wheel on vehicle 102 that was previously having difficulty with obtaining traction now is able to achieve traction, such as the scenario where an icy portion of the road has been maneuvered through, and a non-icy portion of the road exists for a wheel.

Server 108, having received the updated data from vehicle 102 from a message such as an Updated Vehicle Data message 172, may determine updated software based on a current condition of vehicle 102. This software may be a revision of the previous software 160 or may be a further software update. The updated software 174 may be based on a new environment that vehicle 102 has entered after being remotely operated. For example, the original environment may not include an icy surface, yet after being maneuvered remotely, the tires may now encounter an icy surface. The updated software 174 is sent to vehicle 102 processor 104 via a message, such as an Update Vehicle (Updated Software) message 176. Processor 104 may update the vehicle 102 as further disclosed herein, wherein the updated software is installed based on the received updated software in the message. In one embodiment, the original software 164 remains in the vehicle, and the updated software 178 is added to vehicle 102. In another embodiment, the original software 164 is removed, and the updated software 178 is installed. In yet another embodiment, the original software 164 is updated or patched with the updated software 178. Vehicle 102 is continued to be remotely operated 180 by a remote operator associated with server 108, which may be performed by a processor associated with server 108 and/or a remote operator. Communication 182 may occur between processor 104 and server 108 in the remote operation process 180. When the inoperability of the vehicle is below

12

the threshold, processor 104 may request a cessation of the continued remote control 182 of vehicle 102.

In one embodiment, after the updated software is installed 178 on vehicle 102 if the inoperability of vehicle 102 continues, a notification may be sent to proximate vehicles. This notification may occur through V2V communication, for example. The vehicle 102 and/or server 108 may initiate this communication, wherein the proximate vehicles may be determined by one or more of vehicle 102, and server 108. For example, vehicle 102 is having trouble being operated. The system 100 determines a proximate vehicle may be able to assist. The other vehicle may have towing capability, for example, as determined by vehicle data contained in server 108, for example. This other vehicle may be notified to consider assisting vehicle 102.

In one embodiment, the vehicle software 162 and the updated vehicle software 176 may assist in maneuvering of the vehicle from immediate danger in a time that is less than another threshold. For example, if processor 104 determines, based on an analysis of traffic in a proximate area, the threshold may be 15 seconds. The traffic data may be obtained by communication with a remote server, such as a server containing traffic data in the area.

In one embodiment, the cessation of remote operation 182 may occur when the inoperability of vehicle 102 is below the threshold for an amount greater than an amount of time, for a distance, based on weather information, and based on sensor 106 data, or until vehicle 102 comes to a complete stop. For example, if vehicle 102 is operated successfully for over 5 minutes and/or for a distance over 1 mile. The weather information may be obtained by processor 104 by communication with an external server, such as a server containing weather information. For example, if the temperature is below an amount, such as below 27 degrees, and there is either current or forecasted precipitation in the proximate area, the remote operation of vehicle 102 may continue for a greater amount of time due to the possible ice conditions on the road.

In one embodiment, server 108 may perform the actions of determining the inoperability 154 of vehicle 102 through analysis of data 158 received from vehicle 102. In one embodiment, server 108 may determine a time to cease the remote operation of vehicle 102 through analysis of received data 172.

In one embodiment, responsive to an inoperability of vehicle 102 being above a threshold, and a condition of at least one passenger in vehicle 102 being above another threshold, the vehicle may request emergency services. The condition of the at least one passenger may be determined by processor 104 through analysis of received data from sensors 106, such as media including the at least one passenger. For example, the received media includes a passenger that is not sitting upright where previously the passenger was upright or an analysis of the face of the passenger where the analysis determines a level of stress, sickness, and the like. In one embodiment, if the condition of the at least one passenger is worsening over a determined period, and/or an amount of time passes before assistance arrives at vehicle 102 (such as after an accident), system 100 may override the limitations of the current application and request emergency services outright, based on the current or worsening of at least one passenger in vehicle 102.

In an alternate embodiment, a condition may arise where the event may be an object in the road or a change in a condition of the road, such as due to a decrease in visibility and the like. Processor 104 may perform advanced analytics using machine learning (ML) models to assess the situation.

13

This includes identifying a type of obstacle or road condition, assessing the risks, and determining potential determining data related to maneuvers to avoid an obstacle or safely navigate a road condition. These ML models can be pre-trained or can continually learn, such as through communication with an outside entity, such as server **108**.

For example, sensors **106** (such as LIDAR, Radar, ultrasonic sensors, etc.) on vehicle **102** may detect objects, changes in the road, debris in the road, etc. Data from the sensors **106** may be sent to processor **104** for analysis. The processor may utilize advanced software to filter and interpret the data. An ML model may be utilized via processor **104**, wherein the received data is analyzed, which may identify and classify objects detected in the received data, such as objects on the road. For example, the ML model could recognize that there is an object on the road and classify it as a stationary vehicle, pedestrian, debris, etc. Additionally, the ML model may detect a condition of the road, such as an icy environment, the road being wet, or low visibility due to fog in the environment. Processor **104** may assess a level of risk associated with the determined situation. For example, if an object is detected, processor **104** may estimate the likelihood of a collision based on the vehicle's speed and trajectory. In case of poor road conditions, it can estimate how the current conditions may affect the vehicle's handling and safety. The ML model can be adaptive or can learn from new data. In another embodiment, processor **104** may send data back to an outside entity, such as server **108**, for analysis. This server may be part of a larger network where data from multiple vehicles is aggregated. The ML models can be retrained with this aggregated data to improve accuracy and performance. After retraining the ML models with new data, the updated ML models can be sent back to the vehicles.

In an alternate embodiment, vehicle **102** may utilize V2X (Vehicle-to-Everything) communication to communicate with proximate vehicles, infrastructure, and devices to gather additional information about the environment and to provide cooperative awareness. This could include, for example, receiving data from a nearby vehicle that has already navigated the obstacle or road condition or from a processor that is roadside, containing weather information, for example. Processor **104** may aggregate data from sensors **106**, telematics, and V2X communication, including this data in messages sent to server **108**, such as messages **158** and **172**. Processor **104** may send a query message to processors associated with other proximate devices, such as other proximate vehicles, devices associated with proximate infrastructure, and the like. Response messages to the query may be received at processor **104** and allow processor **104** to increase awareness of the environment, traffic related to the environment, debris in the road such as through received media data, and the like.

Flow diagrams depicted herein, such as FIG. 1A, FIG. 1B, FIG. 1C, FIG. 2C, FIG. 2D, FIG. 2E, FIG. 3A, FIG. 3B and FIG. 3C, are separate examples but may be the same or different embodiments. Any of the operations in one flow diagram could be adopted and shared with another flow diagram. No example operation is intended to limit the subject matter of any embodiment or corresponding claim.

It is important to note that all the flow diagrams and corresponding processes derived from FIG. 1A, FIG. 1B, FIG. 1C, FIG. 2C, FIG. 2D, FIG. 2E, FIG. 3A, FIG. 3B and FIG. 3C may be part of a same process or may share sub-processes with one another thus making the diagrams combinable into a single preferred embodiment that does not require any one specific operation but which performs

14

certain operations from one example process and from one or more additional processes. All the example processes are related to the same physical system and can be used separately or interchangeably.

The instant solution can be used in conjunction with one or more types of vehicles: battery electric vehicles, hybrid vehicles, fuel cell vehicles, internal combustion engine vehicles and/or vehicles utilizing renewable sources.

FIG. 2A illustrates a transport network diagram **200**, according to example embodiments. The network comprises elements including a transport **202** including a processor **204**, as well as a transport **202'** including a processor **204'**. The transports **202**, **202'** communicate with one another via the processors **204**, **204'**, as well as other elements (not shown) including transceivers, transmitters, receivers, storage, sensors, and other elements capable of providing communication. The communication between the transports **202**, and **202'** can occur directly, via a private and/or a public network (not shown), or via other transports and elements comprising one or more of a processor, memory, and software. Although depicted as single transports and processors, a plurality of transports and processors may be present. One or more of the applications, features, steps, solutions, etc., described and/or depicted herein may be utilized and/or provided by the instant elements.

FIG. 2B illustrates another transport network diagram **210**, according to example embodiments. The network comprises elements including a transport **202** including a processor **204**, as well as a transport **202'** including a processor **204'**. The transports **202**, **202'** communicate with one another via the processors **204**, **204'**, as well as other elements (not shown), including transceivers, transmitters, receivers, storage, sensors, and other elements capable of providing communication. The communication between the transports **202**, and **202'** can occur directly, via a private and/or a public network (not shown), or via other transports and elements comprising one or more of a processor, memory, and software. The processors **204**, **204'** can further communicate with one or more elements **230** including sensor **212**, wired device **214**, wireless device **216**, database **218**, mobile phone **220**, transport **222**, computer **224**, I/O device **226**, and voice application **228**. The processors **204**, **204'** can further communicate with elements comprising one or more of a processor, memory, and software.

Although depicted as single transports, processors and elements, a plurality of transports, processors and elements may be present. Information or communication can occur to and/or from any of the processors **204**, **204'** and elements **230**. For example, the mobile phone **220** may provide information to the processor **204**, which may initiate the transport **202** to take an action, may further provide the information or additional information to the processor **204'**, which may initiate the transport **202'** to take an action, may further provide the information or additional information to the mobile phone **220**, the transport **222**, and/or the computer **224**. One or more of the applications, features, steps, solutions, etc., described and/or depicted herein may be utilized and/or provided by the instant elements.

FIG. 2C illustrates yet another transport network diagram **240**, according to example embodiments. The network comprises elements including a transport **202**, a processor **204**, and a non-transitory computer-readable medium **242C**. The processor **204** is communicably coupled to the computer-readable medium **242C** and elements **230** (which were depicted in FIG. 2B). The transport **202** could be a transport, server, or any device with a processor and memory.

15

The processor **204** performs one or more of responsive to an inoperability of a vehicle being above a threshold, requesting for the vehicle to be remotely operated **244C**, receiving software, by the vehicle, based on the level of inoperability **246C**, responsive to an installation of the software on the vehicle, remotely operating the vehicle using the installed software **248C**, responsive to the inoperability of the vehicle above the threshold while it is remotely operated, receiving an update to the software **250C**, responsive to an installation of the updated software on the vehicle, further remotely operating the vehicle using the installed updated software **252C**, and responsive to the inoperability below the threshold, requesting a cessation of the further remote operation of the vehicle **252C**.

FIG. 2D illustrates a further transport network diagram **250**, according to example embodiments. The network comprises elements including a transport **202** a processor **204**, and a non-transitory computer-readable medium **242D**. The processor **204** is communicably coupled to the computer-readable medium **242D** and elements **230** (which were depicted in FIG. 2B). The transport **202** could be a transport, server or any device with a processor and memory.

The processor **204** performs one or more of one or more of the software and the updated software include an enhancement of functionality that can be performed by the vehicle and assists in maneuvering the vehicle **244D**, one or more of the software and the updated software assists in maneuvering the vehicle from immediate danger in a time that is less than a threshold **245D**, responsive to one or more of the remote operating of the vehicle, and the further remote operating of the vehicle, sending data to a server, wherein the data is related to a condition of the vehicle **246D**, the updated software is based on a new environmental condition of the vehicle after the remote operating, when the new environment is different from an original environmental condition **247D**, responsive to the inoperability being below the threshold after the updated software is installed on the vehicle, sending a notification of the inoperability to proximate vehicles **248D**, responsive to an inoperability of a vehicle being above a threshold and a condition of at least one passenger of the vehicle being above another threshold, requesting, by the vehicle, emergency services **249D**.

FIG. 2E illustrates yet a further transport network diagram **260**, according to example embodiments. Referring to FIG. 2E, the network diagram **260** includes a transport **202** connected to other transports **202'** and to an update server node **203** over a blockchain network **206**. The transports **202** and **202'** may represent transports/vehicles. The blockchain network **206** may have a ledger **208** for storing software update validation data and a source **207** of the validation for future use (e.g., for an audit).

While this example describes in detail only one transport **202**, multiple such nodes may be connected to the blockchain **206**. It should be understood that the transport **202** may include additional components and that some of the components described herein may be removed and/or modified without departing from a scope of the instant application. The transport **202** may have a computing device or a server computer, or the like, and may include a processor **204**, which may be a semiconductor-based microprocessor, a central processing unit (CPU), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), and/or another hardware device. Although a single processor **204** is depicted, it should be understood that the transport **202** may include multiple processors, multiple cores, or the like without departing from the scope of the

16

instant application. The transport **202** could be a transport, server or any device with a processor and memory.

The processor **204** performs one or more of receiving a confirmation of an event from one or more elements described or depicted herein, wherein the confirmation comprises a blockchain consensus between peers represented by any of the elements **244E** and executing a smart contract to record the confirmation on a blockchain-based on the blockchain consensus **246E**. Consensus is formed between one or more of any element **230** and/or any element described or depicted herein, including a transport, a server, a wireless device, etc. In another example, the transport **202** can be one or more of any element **230** and/or any element described or depicted herein, including a server, a wireless device, etc.

The processors and/or computer-readable medium **242E** may fully or partially reside in the interior or exterior of the transports. The steps or features stored in the computer-readable medium **242E** may be fully or partially performed by any of the processors and/or elements in any order. Additionally, one or more steps or features may be added, omitted, combined, performed at a later time, etc.

FIG. 2F illustrates a diagram **265** depicting the electrification of one or more elements. In one example, a transport **266** may provide power stored in its batteries to one or more elements, including other transport(s) **268**, charging station(s) **270**, and electric grid(s) **272**. The electric grid(s) **272** is/are coupled to one or more of the charging stations **270**, which may be coupled to one or more of the transports **268**. This configuration allows the distribution of electricity/power received from the transport **266**. The transport **266** may also interact with the other transport(s) **268**, such as via Vehicle to Vehicle (V2V) technology, communication over cellular, WiFi, and the like. The transport **266** may also interact wirelessly and/or wired with other transports **268**, the charging station(s) **270** and/or with the electric grid(s) **272**. In one example, the transport **266** is routed (or routes itself) in a safe and efficient manner to the electric grid(s) **272**, the charging station(s) **270**, or the other transport(s) **268**. Using one or more embodiments of the instant solution, the transport **266** can provide energy to one or more of the elements depicted herein in various advantageous ways as described and/or depicted herein. Further, the safety and efficiency of the transport may be increased, and the environment may be positively affected as described and/or depicted herein.

The term 'energy', 'electricity', 'power', and the like may be used to denote any form of energy received, stored, used, shared, and/or lost by the vehicles(s). The energy may be referred to in conjunction with a voltage source and/or a current supply of charge provided from an entity to the transport(s) during a charge/use operation. Energy may also be in the form of fossil fuels (for example, for use with a hybrid transport) or via alternative power sources, including but not limited to lithium-based, nickel-based, hydrogen fuel cells, atomic/nuclear energy, fusion-based energy sources, and energy generated on-the-fly during an energy sharing and/or usage operation for increasing or decreasing one or more transports energy levels at a given time.

In one example, the charging station **270** manages the amount of energy transferred from the transport **266** such that there is sufficient charge remaining in the transport **266** to arrive at a destination. In one example, a wireless connection is used to wirelessly direct an amount of energy transfer between transports **268**, wherein the transports may both be in motion. In one embodiment, wireless charging may occur via a fixed charger and batteries of the transport in alignment with one another (such as a charging mat in a

garage or parking space). In one example, an idle vehicle, such as a vehicle 266 (which may be autonomous) is directed to provide an amount of energy to a charging station 270 and return to the original location (for example, its original location or a different destination). In one example, a mobile energy storage unit (not shown) is used to collect surplus energy from at least one other transport 268 and transfer the stored surplus energy at a charging station 270. In one example, factors determine an amount of energy to transfer to a charging station 270, such as distance, time, as well as traffic conditions, road conditions, environmental/ weather conditions, the vehicle's condition (weight, etc.), an occupant(s) schedule while utilizing the vehicle, a prospective occupant(s) schedule waiting for the vehicle, etc. In one example, the transport(s) 268, the charging station(s) 270 and/or the electric grid(s) 272 can provide energy to the transport 266.

In one embodiment, a location such as a building, a residence, or the like (not depicted), communicably coupled to one or more of the electric grid 272, the transport 266, and/or the charging station(s) 270. The rate of electric flow to one or more of the location, the transport 266, the other transport(s) 268 is modified, depending on external conditions, such as weather. For example, when the external temperature is extremely hot or extremely cold, raising the chance for an outage of electricity, the flow of electricity to a connected vehicle 266/268 is slowed to help minimize the chance for an outage.

In one embodiment, transports 266 and 268 may be utilized as bidirectional transports. Bidirectional transports are those that may serve as mobile microgrids that can assist in the supplying of electrical power to the grid 272 and/or reduce the power consumption when the grid is stressed. Bidirectional transports incorporate bidirectional charging, which in addition to receiving a charge to the transport, the transport can take energy from the transport and "push" the energy back into the grid 272, otherwise referred to as "V2G". In bidirectional charging, the electricity flows both ways; to the transport and from the transport. When a transport is charged, alternating current (AC) electricity from the grid 272 is converted to direct current (DC). This may be performed by one or more of the transport's own converter or a converter on the charger 270. The energy stored in the transport's batteries may be sent in an opposite direction back to the grid. The energy is converted from DC to AC through a converter usually located in the charger 270, otherwise referred to as a bidirectional charger. Further, the instant solution as described and depicted with respect to FIG. 2F can be utilized in this and other networks and/or systems.

FIG. 2G is a diagram showing interconnections between different elements 275. The instant solution may be stored and/or executed entirely or partially on and/or by one or more computing devices 278', 279', 281', 282', 283', 284', 276', 285', 287' and 277' associated with various entities, all communicably coupled and in communication with a network 286. A database 287 is communicably coupled to the network and allows for the storage and retrieval of data. In one example, the database is an immutable ledger. One or more of the various entities may be a transport 276, one or more service provider 279, one or more public buildings 281, one or more traffic infrastructure 282, one or more residential dwellings 283, an electric grid/charging station 284, a microphone 285, and/or another transport 277. Other entities and/or devices, such as one or more private users using a smartphone 278, a laptop 280, an augmented reality (AR) device, a virtual reality (VR) device, and/or any

wearable device may also interwork with the instant solution. The smartphone 278, laptop 280, the microphone 285, and other devices may be connected to one or more of the connected computing devices 278', 279', 281', 282', 283', 284', 276', 285', 287', and 277'. The one or more public buildings 281 may include various agencies. The one or more public buildings 281 may utilize a computing device 281'. The one or more service provider 279 may include a dealership, a tow truck service, a collision center or other repair shop. The one or more service provider 279 may utilize a computing apparatus 279'. These various computer devices may be directly and/or communicably coupled to one another, such as via wired networks, wireless networks, blockchain networks, and the like. The microphone 285 may be utilized as a virtual assistant, in one example. In one example, the one or more traffic infrastructure 282 may include one or more traffic signals, one or more sensors including one or more cameras, vehicle speed sensors or traffic sensors, and/or other traffic infrastructure. The one or more traffic infrastructure 282 may utilize a computing device 282'.

In one embodiment, anytime an electrical charge is given or received to/from a charging station and/or an electrical grid, the entities that allow that to occur are one or more of a vehicle, a charging station, a server, and a network communicably coupled to the vehicle, the charging station, and the electrical grid.

In one example, a transport 277/276 can transport a person, an object, a permanently or temporarily affixed apparatus, and the like. In one example, the transport 277 may communicate with transport 276 via V2V communication through the computers associated with each transport 276' and 277' and may be referred to as a transport, car, vehicle, automobile, and the like. The transport 276/277 may be a self-propelled wheeled conveyance, such as a car, a sports utility vehicle, a truck, a bus, a van, or other motor or battery-driven or fuel cell-driven transport. For example, transport 276/277 may be an electric vehicle, a hybrid vehicle, a hydrogen fuel cell vehicle, a plug-in hybrid vehicle, or any other type of vehicle with a fuel cell stack, a motor, and/or a generator. Other examples of vehicles include bicycles, scooters, trains, planes, boats, and any other form of conveyance that is capable of transportation. The transport 276/277 may be semi-autonomous or autonomous. For example, transport 276/277 may be self-manuevering and navigate without human input. An autonomous vehicle may have and use one or more sensors and/or a navigation unit to drive autonomously.

FIG. 2H is another block diagram showing interconnections between different elements in one example 290. A transport 276 is presented and includes ECUs 295, 296, and a Head Unit (otherwise known as an Infotainment System) 297. An Electrical Control Unit (ECU) is an embedded system in automotive electronics controlling one or more of the electrical systems or subsystems in a transport. ECUs may include but are not limited to the management of a transport's engine, brake system, gearbox system, door locks, dashboard, airbag system, infotainment system, electronic differential, and active suspension. ECUs are connected to the transport's Controller Area Network (CAN) bus 294. The ECUs may also communicate with a transport computer 298 via the CAN bus 294. The transport's processors/sensors (such as the transport computer) 298 can communicate with external elements, such as a server 293 via a network 292 (such as the Internet). Each ECU 295, 296, and Head Unit 297 may contain its own security policy. The security policy defines permissible processes that can be

executed in the proper context. In one example, the security policy may be partially or entirely provided in the transport computer **298**.

ECUs **295**, **296**, and Head Unit **297** may each include a custom security functionality element **299** defining authorized processes and contexts within which those processes are permitted to run. Context-based authorization to determine validity if a process can be executed allows ECUs to maintain secure operation and prevent unauthorized access from elements such as the transport's Controller Area Network (CAN Bus). When an ECU encounters a process that is unauthorized, that ECU can block the process from operating. Automotive ECUs can use different contexts to determine whether a process is operating within its permitted bounds, such as proximity contexts such as nearby objects, distance to approaching objects, speed, and trajectory relative to other moving objects, and operational contexts such as an indication of whether the transport is moving or parked, the transport's current speed, the transmission state, user-related contexts such as devices connected to the transport via wireless protocols, use of the infotainment, cruise control, parking assist, driving assist, location-based contexts, and/or other contexts.

Referring to FIG. 2I, an operating environment **290A** for a connected transport, is illustrated according to some embodiments. As depicted, the transport **276** includes a Controller Area Network (CAN) bus **291A** connecting elements **292A-299A** of the transport. Other elements may be connected to the CAN bus and are not depicted herein. The depicted elements connected to the CAN bus include a sensor set **292A**, Electronic Control Units **293A**, autonomous features or Advanced Driver Assistance Systems (ADAS) **294A**, and the navigation system **295A**. In some embodiments, the transport **276** includes a processor **296A**, a memory **297A**, a communication unit **298A**, and an electronic display **299A**.

The processor **296A** includes an arithmetic logic unit, a microprocessor, a general-purpose controller, and/or a similar processor array to perform computations and provide electronic display signals to a display unit **299A**. The processor **296A** processes data signals and may include various computing architectures, including a complex instruction set computer (CISC) architecture, a reduced instruction set computer (RISC) architecture, or an architecture implementing a combination of instruction sets. The transport **276** may include one or more processors **296A**. Other processors, operating systems, sensors, displays, and physical configurations that are communicably coupled to one another (not depicted) may be used with the instant solution.

Memory **297A** is a non-transitory memory storing instructions or data that may be accessed and executed by the processor **296A**. The instructions and/or data may include code to perform the techniques described herein. The memory **297A** may be a dynamic random-access memory (DRAM) device, a static random-access memory (SRAM) device, flash memory, or another memory device. In some embodiments, the memory **297A** also may include non-volatile memory or a similar permanent storage device and media, which may include a hard disk drive, a floppy disk drive, a CD-ROM device, a DVD-ROM device, a DVD-RAM device, a DVD-RW device, a flash memory device, or some other mass storage device for storing information on a permanent basis. A portion of the memory **297A** may be reserved for use as a buffer or virtual random-access

memory (virtual RAM). The transport **276** may include one or more memories **297A** without deviating from the current solution.

The memory **297A** of the transport **276** may store one or more of the following types of data: navigation route data **295A**, and autonomous features data **294A**. In some embodiments, the memory **297A** stores data that may be necessary for the navigation application **295A** to provide the functions.

The navigation system **295A** may describe at least one navigation route including a start point and an endpoint. In some embodiments, the navigation system **295A** of the transport **276** receives a request from a user for navigation routes wherein the request includes a starting point and an ending point. The navigation system **295A** may query a real-time data server **293** (via a network **292**), such as a server that provides driving directions, for navigation route data corresponding to navigation routes, including the start point and the endpoint. The real-time data server **293** transmits the navigation route data to the transport **276** via a wireless network **292**, and the communication system **298A** stores the navigation data **295A** in the memory **297A** of the transport **276**.

The ECU **293A** controls the operation of many of the systems of the transport **276**, including the ADAS systems **294A**. The ECU **293A** may, responsive to instructions received from the navigation system **295A**, deactivate any unsafe and/or unselected autonomous features for the duration of a journey controlled by the ADAS systems **294A**. In this way, the navigation system **295A** may control whether ADAS systems **294A** are activated or enabled so that they may be activated for a given navigation route.

The sensor set **292A** may include any sensors in the transport **276** generating sensor data. For example, the sensor set **292A** may include short-range sensors and long-range sensors. In some embodiments, the sensor set **292A** of the transport **276** may include one or more of the following vehicle sensors: a camera, a Lidar sensor, an ultrasonic sensor, an automobile engine sensor, a radar sensor, a laser altimeter, a manifold absolute pressure sensor, an infrared detector, a motion detector, a thermostat, a sound detector, a carbon monoxide sensor, a carbon dioxide sensor, an oxygen sensor, a mass airflow sensor, an engine coolant temperature sensor, a throttle position sensor, a crankshaft position sensor, a valve timer, an air-fuel ratio meter, a blind spot meter, a curb feeler, a defect detector, a Hall effect sensor, a parking sensor, a radar gun, a speedometer, a speed sensor, a tire-pressure monitoring sensor, a torque sensor, a transmission fluid temperature sensor, a turbine speed sensor (TSS), a variable reluctance sensor, a vehicle speed sensor (VSS), a water sensor, a wheel speed sensor, a GPS sensor, a mapping functionality, and any other type of automotive sensor. The navigation system **295A** may store the sensor data in the memory **297A**.

The communication unit **298A** transmits and receives data to and from the network **292** or to another communication channel. In some embodiments, the communication unit **298A** may include a DSRC transceiver, a DSRC receiver, and other hardware or software necessary to make the transport **276** a DSRC-equipped device.

The transport **276** may interact with other transports **277** via V2V technology. V2V communication includes sensing radar information corresponding to relative distances to external objects, receiving GPS information of the transports, setting areas as areas where the other transports **277** are located based on the sensed radar information, calculating probabilities that the GPS information of the object vehicles will be located at the set areas, and identifying

21

transports and/or objects corresponding to the radar information and the GPS information of the object vehicles based on the calculated probabilities, in one example.

For a transport to be adequately secured, the transport must be protected from unauthorized physical access as well as unauthorized remote access (e.g., cyber-threats). To prevent unauthorized physical access, a transport is equipped with a secure access system such as a keyless entry in one example. Meanwhile, security protocols are added to a transport's computers and computer networks to facilitate secure remote communications to and from the transport in one example.

Electronic Control Units (ECUs) are nodes within a transport that control tasks such as activating the windshield wipers to tasks such as an anti-lock brake system. ECUs are often connected to one another through the transport's central network, which may be referred to as a controller area network (CAN). State-of-the-art features such as autonomous driving are strongly reliant on implementing new, complex ECUs such as advanced driver-assistance systems (ADAS), sensors, and the like. While these new technologies have helped improve the safety and driving experience of a transport, they have also increased the number of externally-communicating units inside of the transport, making them more vulnerable to attack. Below are some examples of protecting the transport from physical intrusion and remote intrusion.

FIG. 2J illustrates a keyless entry system 290B to prevent unauthorized physical access to a transport 291B, according to example embodiments. Referring to FIG. 2J, a key fob 292B transmits commands to a transport 291B using radio frequency signals in one example. In this example, the key fob 292B includes a transmitter 2921B with an antenna that is capable of sending short-range wireless radio signals. The transport 291B includes a receiver 2911B with an antenna that is capable of receiving the short-range wireless signal transmitted from the transmitter 2921B. The key fob 292B and the transport 291B also include CPUs 2922B and 2913B, respectively, which control the respective devices. Here, a memory of the CPUs 2922B and 2913B (or accessible to the CPUs). Each of the key fob 292B and the transport 291B includes power supplies 2924B and 2915B for powering the respective devices in one example.

When the user presses a button 293B (or otherwise actuates the fob, etc.) on the key fob 292B, the CPU 2922B wakes up inside the key fob 292B and sends a data stream to the transmitter 2921B, which is output via the antenna. In other embodiments, the user's intent is acknowledged on the key fob 292B via other means, such as via a microphone that accepts audio, a camera that captures images and/or video, or other sensors that are commonly utilized in the art to detect intent from a user including receiving gestures, motion, eye movements, and the like. The data stream may be a 64-bit to 128-bit long signal, which includes one or more of a preamble, a command code, and a rolling code. The signal may be sent at a rate between 2 KHz and 20 KHz, but embodiments are not limited thereto. In response, the receiver 2911B of the transport 291B captures the signal from the transmitter 2921B, demodulates the signal, and sends the data stream to the CPU 2913B, which decodes the signal and sends commands (e.g., lock the door, unlock the door, etc.) to a command module 2912B.

If the key fob 292B and the transport 291B use a fixed code between them, replay attacks can be performed. In this case, if the attacker can capture/sniff the fixed code during the short-range communication, the attacker could replay this code to gain entry into the transport 291B. To improve

22

security, the key fob and the transport 291B may use a rolling code that changes after each use. Here, the key fob 292B and the transport 291B are synchronized with an initial seed 2923B (e.g., a random number, pseudo-random number, etc.) This is referred to as pairing. The key fob 292B and the transport 291B also include a shared algorithm for modifying the initial seed 2914B each time the button 293B is pressed. The following keypress will take the result of the previous keypress as an input and transform it into the next number in the sequence. In some cases, the transport 291B may store multiple next codes (e.g., 255 next codes) in case the keypress on the key fob 292B is not detected by the transport 291B. Thus, a number of keypress on the key fob 292B that are unheard by the transport 291B do not prevent the transport from becoming out of sync.

In addition to rolling codes, the key fob 292B and the transport 291B may employ other methods to make attacks even more difficult. For example, different frequencies may be used for transmitting the rolling codes. As another example, two-way communication between the transmitter 2921B and the receiver 2911B may be used to establish a secure session. As another example, codes may have limited expirations or timeouts. Further, the instant solution as described and depicted with respect to FIG. 2J can be utilized in this and other networks and/or systems, including those that are described and depicted herein.

FIG. 2K illustrates a controller area network (CAN) 290C within a transport, according to example embodiments. Referring to FIG. 2K, the CAN 290C includes a CAN bus 297C with a high and low terminal and a plurality of electronic control units (ECUs) 291C, 292C, 293C, etc. which are connected to the CAN bus 297C via wired connections. The CAN bus 297C is designed to allow microcontrollers and devices to communicate with each other in an application without a host computer. The CAN bus 297C implements a message-based protocol (i.e., ISO 11898 standards) that allows ECUs 291C-293C to send commands to one another at a root level. Meanwhile, the ECUs 291C-293C represent controllers for controlling electrical systems or subsystems within the transport. Examples of the electrical systems include power steering, anti-lock brakes, air-conditioning, tire pressure monitoring, cruise control, and many other features.

In this example, the ECU 291C includes a transceiver 2911C and a microcontroller 2912C. The transceiver may be used to transmit and receive messages to and from the CAN bus 297C. For example, the transceiver 2911C may convert the data from the microcontroller 2912C into a format of the CAN bus 297C and also convert data from the CAN bus 297C into a format for the microcontroller 2912C. Meanwhile, the microcontroller 2912C interprets the messages and also decide what messages to send using ECU software installed therein in one example.

To protect the CAN 290C from cyber threats, various security protocols may be implemented. For example, sub-networks (e.g., sub-networks A and B, etc.) may be used to divide the CAN 290C into smaller sub-CANs and limit an attacker's capabilities to access the transport remotely. In the example of FIG. 2K, ECUs 291C and 292C may be part of a same sub-network, while ECU 293C is part of an independent sub-network. Furthermore, a firewall 294C (or gateway, etc.) may be added to block messages from crossing the CAN bus 297C across sub-networks. If an attacker gains access to one sub-network, the attacker will not have access to the entire network. To make sub-networks even more secure, the most critical ECUs are not placed on the same sub-network, in one example.

Although not shown in FIG. 2K, other examples of security controls within a CAN include an intrusion detection system (IDS) which can be added to each sub-network and read all data passing to detect malicious messages. If a malicious message is detected, the IDS can notify the automobile user. Other possible security protocols include encryption/security keys that can be used to obscure messages. As another example, authentication protocols are implemented that enables a message to authenticate itself, in one example.

In addition to protecting a transport's internal network, transports may also be protected when communicating with external networks such as the Internet. One of the benefits of having a transport connection to a data source such as the Internet is that information from the transport can be sent through a network to remote locations for analysis. Examples of transport information include GPS, onboard diagnostics, tire pressure, and the like. These communication systems are often referred to as telematics because they involve the combination of telecommunications and informatics. Further, the instant solution as described and depicted with respect to FIG. 2K can be utilized in this and other networks and/or systems, including those that are described and depicted herein.

FIG. 2L illustrates a secure end-to-end transport communication channel according to example embodiments. Referring to FIG. 2L, a telematics network 290D includes a transport 291D and a host server 295D that is disposed at a remote location (e.g., a web server, a cloud platform, a database, etc.) and connected to the transport 291D via a network such as the Internet. In this example, a device 296D associated with the host server 295D may be installed within the network inside the transport 291D. Furthermore, although not shown, the device 296D may connect to other elements of the transport 291D, such as the CAN bus, an onboard diagnostics (OBDII) port, a GPS system, a SIM card, a modem, and the like. The device 296D may collect data from any of these systems and transfer the data to the server 295D via the network.

Secure management of data begins with the transport 291D. In some embodiments, the device 296D may collect information before, during, and after a trip. The data may include GPS data, travel data, passenger information, diagnostic data, fuel data, speed data, and the like. However, the device 296D may only communicate the collected information back to the host server 295D in response to transport ignition and trip completion. Furthermore, communication may only be initiated by the device 296D and not by the host server 295D. As such, the device 296D will not accept communications initiated by outside sources in one example.

To perform the communication, the device 296D may establish a secured private network between the device 296D and the host server 295D. Here, the device 296D may include a tamper-proof SIM card that provides secure access to a carrier network 294D via a radio tower 292D. When preparing to transmit data to the host server 295D, the device 296D may establish a one-way secure connection with the host server 295D. The carrier network 294D may communicate with the host server 295D using one or more security protocols. As a non-limiting example, the carrier network 294D may communicate with the host server 295D via a VPN tunnel which allows access through a firewall 293D of the host server 295D. As another example, the carrier network 294D may use data encryption (e.g., AES encryption, etc.) when transmitting data to the host server 295D. In

some cases, the system may use multiple security measures such as both a VPN and encryption to further secure the data.

In addition to communicating with external servers, transports may also communicate with each other. In particular, transport-to-transport (V2V) communication systems enable transports to communicate with each other, roadside infrastructures (e.g., traffic lights, signs, cameras, parking meters, etc.), and the like, over a wireless network. The wireless network may include one or more of Wi-Fi networks, cellular networks, dedicated short-range communication (DSRC) networks, and the like. Transports may use V2V communication to provide other transports with information about a transport's speed, acceleration, braking, and direction, to name a few. Accordingly, transports can receive insight into the conditions ahead before such conditions become visible, thus greatly reducing collisions. Further, the instant solution as described and depicted with respect to FIG. 2L can be utilized in this and other networks and/or systems, including those that are described and depicted herein.

FIG. 2M illustrates an example 290E of transports 293E and 292E performing secured V2V communications using security certificates, according to example embodiments. Referring to FIG. 2M, the transports 293E and 292E may communicate via V2V communications over a short-range network, a cellular network, or the like. Before sending messages, the transports 293E and 292E may sign the messages using a respective public key certificate. For example, the transport 293E may sign a V2V message using a public key certificate 294E. Likewise, the transport 292E may sign a V2V message using a public key certificate 295E. The public key certificates 294E and 295E are associated with the transports 293E and 292E, respectively, in one example.

Upon receiving the communications from each other, the transports may verify the signatures with a certificate authority 291E or the like. For example, the transport 292E may verify with the certificate authority 291E that the public key certificate 294E used by transport 293E to sign a V2V communication is authentic. If the transport 292E successfully verifies the public key certificate 294E, the transport knows that the data is from a legitimate source. Likewise, the transport 293E may verify with the certificate authority 291E that the public key certificate 295E used by the transport 292E to sign a V2V communication is authentic. Further, the instant solution as described and depicted with respect to FIG. 2M can be utilized in this and other networks and/or systems including those that are described and depicted herein.

FIG. 2N illustrates yet a further diagram 290F depicting an example of a transport interacting with a security processor and a wireless device, according to example embodiments. In some embodiments, the computer 224 shown in FIG. 2B may include security processor 292F as shown in the process 290F of the example of FIG. 2N. In particular, the security processor 292F may perform authorization, authentication, cryptography (e.g., encryption), and the like, for data transmissions that are sent between ECUs and other devices on a CAN bus of a vehicle, and also data messages that are transmitted between different vehicles.

In the example of FIG. 2N, the security processor 292F may include an authorization module 293F, an authentication module 294F, and a cryptography module 295F. The security processor 292F may be implemented within the transport's computer and may communicate with other transport elements, for example, the ECUs/CAN network 296F, wired and wireless devices 298F such as wireless

25

network interfaces, input ports, and the like. The security processor **292F** may ensure that data frames (e.g., CAN frames, etc.) that are transmitted internally within a transport (e.g., via the ECUs/CAN network **296F**) are secure. Likewise, the security processor **292F** can ensure that messages transmitted between different transports and devices attached or connected via a wire to the transport's computer are also secured.

For example, the authorization module **293F** may store passwords, usernames, PIN codes, biometric scans, and the like for different transport users. The authorization module **293F** may determine whether a user (or technician) has permission to access certain settings such as a transport's computer. In some embodiments, the authorization module may communicate with a network interface to download any necessary authorization information from an external server. When a user desires to make changes to the transport settings or modify technical details of the transport via a console or GUI within the transport or via an attached/connected device, the authorization module **293F** may require the user to verify themselves in some way before such settings are changed. For example, the authorization module **293F** may require a username, a password, a PIN code, a biometric scan, a predefined line drawing or gesture, and the like. In response, the authorization module **293F** may determine whether the user has the necessary permissions (access, etc.) being requested.

The authentication module **294F** may be used to authenticate internal communications between ECUs on the CAN network of the vehicle. As an example, the authentication module **294F** may provide information for authenticating communications between the ECUS. As an example, the authentication module **294F** may transmit a bit signature algorithm to the ECUs of the CAN network. The ECUs may use the bit signature algorithm to insert authentication bits into the CAN fields of the CAN frame. All ECUs on the CAN network typically receive each CAN frame. The bit signature algorithm may dynamically change the position, amount, etc., of authentication bits each time a new CAN frame is generated by one of the ECUs. The authentication module **294F** may also provide a list of ECUs that are exempt (safe list) and that do not need to use the authentication bits. The authentication module **294F** may communicate with a remote server to retrieve updates to the bit signature algorithm and the like.

The encryption module **295F** may store asymmetric key pairs to be used by the transport to communicate with other external user devices and transports. For example, the encryption module **295F** may provide a private key to be used by the transport to encrypt/decrypt communications, while the corresponding public key may be provided to other user devices and transports to enable the other devices to decrypt/encrypt the communications. The encryption module **295F** may communicate with a remote server to receive new keys, updates to keys, keys of new transports, users, etc., and the like. The encryption module **295F** may also transmit any updates to a local private/public key pair to the remote server.

FIG. 3A illustrates a flow diagram **300**, according to example embodiments. Referring to FIG. 3A, the solution includes one or more of responsive to an inoperability of a vehicle being above a threshold, requesting for the vehicle to be remotely operated **302**, receiving software, by the vehicle, based on the level of inoperability **304**, responsive to an installation of the software on the vehicle, remotely operating the vehicle using the installed software **306**, responsive to the inoperability of the vehicle above the

26

threshold while it is remotely operated, receiving an update to the software **308**, responsive to an installation of the updated software on the vehicle, further remotely operating the vehicle using the installed updated software **310**, and responsive to the inoperability below the threshold, requesting a cessation of the further remote operation of the vehicle **312**.

FIG. 3B illustrates another flow diagram **320**, according to example embodiments. Referring to FIG. 3B, the solution includes one or more of: one or more of the software and the updated software include an enhancement of functionality that can be performed by the vehicle and assists in maneuvering the vehicle **322**, one or more of the software and the updated software assists in maneuvering the vehicle from immediate danger in a time that is less than a threshold **323**, responsive to one or more of the remote operating of the vehicle, and the further remote operating of the vehicle, sending data to a server, wherein the data is related to a condition of the vehicle **324**, the updated software is based on a new environmental condition of the vehicle after the remote operating, when the new environment is different from an original environmental condition **325**, responsive to the inoperability being below the threshold after the updated software is installed on the vehicle, sending a notification of the inoperability to proximate vehicles **326**, responsive to an inoperability of a vehicle being above a threshold and a condition of at least one passenger of the vehicle being above another threshold, requesting, by the vehicle, emergency services **327**.

FIG. 3C illustrates yet another flow diagram **340**, according to example embodiments. Referring to FIG. 3C, the flow diagram includes one or more of receiving a confirmation of an event from one or more elements described or depicted herein, wherein the confirmation comprises a blockchain consensus between peers represented by any of the elements **342** and executing a smart contract to record the confirmation on a blockchain-based on the blockchain consensus **344**.

FIG. 4 illustrates a machine learning transport network diagram **400**, according to example embodiments. The network **400** includes a transport **402** that interfaces with a machine learning subsystem **406**. The transport includes one or more sensors **404**.

The machine learning subsystem **406** contains a learning model **408**, which is an artifact created by a machine learning training system **410** that generates predictions by finding patterns in one or more training data sets. In some embodiments, the machine learning subsystem **406** resides in the transport node **402**. An artifact is used to describe an output created by a training process, such as a checkpoint, a file, or a model. In other embodiments, the machine learning subsystem **406** resides outside of the transport node **402**.

The transport **402** sends data from the one or more sensors **404** to the machine learning subsystem **406**. The machine learning subsystem **406** provides the one or more sensor **404** data to the learning model **408**, which returns one or more predictions. The machine learning subsystem **406** sends one or more instructions to the transport **402** based on the predictions from the learning model **408**.

In a further embodiment, the transport **402** may send the one or more sensor **404** data to the machine learning training system **410**. In yet another example, the machine learning subsystem **406** may send the sensor **404** data to the machine learning subsystem **410**. One or more of the applications, features, steps, solutions, etc., described and/or depicted herein may utilize the machine learning network **400** as described herein.

FIG. 5A illustrates an example vehicle configuration **500** for managing database transactions associated with a vehicle, according to example embodiments. Referring to FIG. 5A, as a particular transport/vehicle **525** is engaged in transactions (e.g., vehicle service, dealer transactions, delivery/pickup, transportation services, etc.), the vehicle may receive assets **510** and/or expel/transfer assets **512** according to a transaction(s). A transport processor **526** resides in the vehicle **525** and communication exists between the transport processor **526**, a database **530**, a transport processor **526** and the transaction module **520**. The transaction module **520** may record information, such as assets, parties, credits, service descriptions, date, time, location, results, notifications, unexpected events, etc. Those transactions in the transaction module **520** may be replicated into a database **530**. The database **530** can be one of a SQL database, an RDBMS, a relational database, a non-relational database, a blockchain, a distributed ledger, and may be on board the transport, may be off-board the transport, may be accessed directly and/or through a network, or be accessible to the transport.

FIG. 5B illustrates an example vehicle configuration **550** for managing database transactions conducted among various vehicles, according to example embodiments. The vehicle **525** may engage with another vehicle **508** to perform various actions such as to share, transfer, acquire service calls, etc. when the vehicle has reached a status where the services need to be shared with another vehicle. For example, the vehicle **508** may be due for a battery charge and/or may have an issue with a tire and may be in route to pick up a package for delivery. A transport processor **528** resides in the vehicle **508** and communication exists between the transport processor **528**, a database **554**, and the transaction module **552**. The vehicle **508** may notify another vehicle **525**, which is in its network and which operates on its blockchain member service. A transport processor **526** resides in the vehicle **525** and communication exists between the transport processor **526**, a database **530**, the transport processor **526** and a transaction module **520**. The vehicle **525** may then receive the information via a wireless communication request to perform the package pickup from the vehicle **508** and/or from a server (not shown). The transactions are logged in the transaction modules **552** and **520** of both vehicles. The credits are transferred from vehicle **508** to vehicle **525** and the record of the transferred service is logged in the database **530/554** assuming that the blockchains are different from one another, or are logged in the same blockchain used by all members. The database **554** can be one of a SQL database, an RDBMS, a relational database, a non-relational database, a blockchain, a distributed ledger, and may be on board the transport, may be off-board the transport, may be accessible directly and/or through a network.

FIG. 6A illustrates a blockchain architecture configuration **600**, according to example embodiments. Referring to FIG. 6A, the blockchain architecture **600** may include certain blockchain elements, for example, a group of blockchain member nodes **602-606** as part of a blockchain group **610**. In one example embodiment, a permissioned blockchain is not accessible to all parties but only to those members with permissioned access to the blockchain data. The blockchain nodes participate in a number of activities, such as blockchain entry addition and validation process (consensus). One or more of the blockchain nodes may endorse entries based on an endorsement policy and may provide an ordering service for all blockchain nodes. A blockchain node may initiate a blockchain action (such as an authentication) and

seek to write to a blockchain immutable ledger stored in the blockchain, a copy of which may also be stored on the underpinning physical infrastructure.

The blockchain transactions **620** are stored in memory of computers as the transactions are received and approved by the consensus model dictated by the members' nodes. Approved transactions **626** are stored in current blocks of the blockchain and committed to the blockchain via a committal procedure, which includes performing a hash of the data contents of the transactions in a current block and referencing a previous hash of a previous block. Within the blockchain, one or more smart contracts **630** may exist that define the terms of transaction agreements and actions included in smart contract executable application code **632**, such as registered recipients, vehicle features, requirements, permissions, sensor thresholds, etc. The code may be configured to identify whether requesting entities are registered to receive vehicle services, what service features they are entitled/required to receive given their profile statuses and whether to monitor their actions in subsequent events. For example, when a service event occurs and a user is riding in the vehicle, the sensor data monitoring may be triggered, and a certain parameter, such as a vehicle charge level, may be identified as being above/below a particular threshold for a particular period of time, then the result may be a change to a current status, which requires an alert to be sent to the managing party (i.e., vehicle owner, vehicle operator, server, etc.) so the service can be identified and stored for reference. The vehicle sensor data collected may be based on types of sensor data used to collect information about vehicle's status. The sensor data may also be the basis for the vehicle event data **634**, such as a location(s) to be traveled, an average speed, a top speed, acceleration rates, whether there were any collisions, was the expected route taken, what is the next destination, whether safety measures are in place, whether the vehicle has enough charge/fuel, etc. All such information may be the basis of smart contract terms **630**, which are then stored in a blockchain. For example, sensor thresholds stored in the smart contract can be used as the basis for whether a detected service is necessary and when and where the service should be performed.

FIG. 6B illustrates a shared ledger configuration, according to example embodiments. Referring to FIG. 6B, the blockchain logic example **640** includes a blockchain application interface **642** as an API or plug-in application that links to the computing device and execution platform for a particular transaction. The blockchain configuration **640** may include one or more applications, which are linked to application programming interfaces (APIs) to access and execute stored program/application code (e.g., smart contract executable code, smart contracts, etc.), which can be created according to a customized configuration sought by participants and can maintain their own state, control their own assets, and receive external information. This can be deployed as an entry and installed, via appending to the distributed ledger, on all blockchain nodes.

The smart contract application code **644** provides a basis for the blockchain transactions by establishing application code, which when executed causes the transaction terms and conditions to become active. The smart contract **630**, when executed, causes certain approved transactions **626** to be generated, which are then forwarded to the blockchain platform **652**. The platform includes a security/authorization **658**, computing devices, which execute the transaction management **656** and a storage portion **654** as a memory that stores transactions and smart contracts in the blockchain.

The blockchain platform may include various layers of blockchain data, services (e.g., cryptographic trust services, virtual execution environment, etc.), and underpinning physical computer infrastructure that may be used to receive and store new entries and provide access to auditors, which are seeking to access data entries. The blockchain may expose an interface that provides access to the virtual execution environment necessary to process the program code and engage the physical infrastructure. Cryptographic trust services may be used to verify entries such as asset exchange entries and keep information private.

The blockchain architecture configuration of FIGS. 6A and 6B may process and execute program/application code via one or more interfaces exposed, and services provided, by the blockchain platform. As a non-limiting example, smart contracts may be created to execute reminders, updates, and/or other notifications subject to the changes, updates, etc. The smart contracts can themselves be used to identify rules associated with authorization and access requirements and usage of the ledger. For example, the information may include a new entry, which may be processed by one or more processing entities (e.g., processors, virtual machines, etc.) included in the blockchain layer. The result may include a decision to reject or approve the new entry based on the criteria defined in the smart contract and/or a consensus of the peers. The physical infrastructure may be utilized to retrieve any of the data or information described herein.

Within smart contract executable code, a smart contract may be created via a high-level application and programming language, and then written to a block in the blockchain. The smart contract may include executable code that is registered, stored, and/or replicated with a blockchain (e.g., distributed network of blockchain peers). An entry is an execution of the smart contract code, which can be performed in response to conditions associated with the smart contract being satisfied. The executing of the smart contract may trigger a trusted modification(s) to a state of a digital blockchain ledger. The modification(s) to the blockchain ledger caused by the smart contract execution may be automatically replicated throughout the distributed network of blockchain peers through one or more consensus protocols.

The smart contract may write data to the blockchain in the format of key-value pairs. Furthermore, the smart contract code can read the values stored in a blockchain and use them in application operations. The smart contract code can write the output of various logic operations into the blockchain. The code may be used to create a temporary data structure in a virtual machine or other computing platform. Data written to the blockchain can be public and/or can be encrypted and maintained as private. The temporary data that is used/generated by the smart contract is held in memory by the supplied execution environment, then deleted once the data needed for the blockchain is identified.

A smart contract executable code may include the code interpretation of a smart contract, with additional features. As described herein, the smart contract executable code may be program code deployed on a computing network, where it is executed and validated by chain validators together during a consensus process. The smart contract executable code receives a hash and retrieves from the blockchain a hash associated with the data template created by use of a previously stored feature extractor. If the hashes of the hash identifier and the hash created from the stored identifier template data match, then the smart contract executable code sends an authorization key to the requested service. The

smart contract executable code may write to the blockchain data associated with the cryptographic details.

FIG. 6C illustrates a blockchain configuration for storing blockchain transaction data, according to example embodiments. Referring to FIG. 6C, the example configuration 660 provides for the vehicle 662, the user device 664 and a server 666 sharing information with a distributed ledger (i.e., blockchain) 668. The server may represent a service provider entity inquiring with a vehicle service provider to share user profile rating information in the event that a known and established user profile is attempting to rent a vehicle with an established rated profile. The server 666 may be receiving and processing data related to a vehicle's service requirements. As the service events occur, such as the vehicle sensor data indicates a need for fuel/charge, a maintenance service, etc., a smart contract may be used to invoke rules, thresholds, sensor information gathering, etc., which may be used to invoke the vehicle service event. The blockchain transaction data 670 is saved for each transaction, such as the access event, the subsequent updates to a vehicle's service status, event updates, etc. The transactions may include the parties, the requirements (e.g., 18 years of age, service eligible candidate, valid driver's license, etc.), compensation levels, the distance traveled during the event, the registered recipients permitted to access the event and host a vehicle service, rights/permissions, sensor data retrieved during the vehicle event operation to log details of the next service event and identify a vehicle's condition status, and thresholds used to make determinations about whether the service event was completed and whether the vehicle's condition status has changed.

FIG. 6D illustrates blockchain blocks 680 that can be added to a distributed ledger, according to example embodiments, and contents of block structures 682A to 682N. Referring to FIG. 6D, clients (not shown) may submit entries to blockchain nodes to enact activity on the blockchain. As an example, clients may be applications that act on behalf of a requester, such as a device, person or entity to propose entries for the blockchain. The plurality of blockchain peers (e.g., blockchain nodes) may maintain a state of the blockchain network and a copy of the distributed ledger. Different types of blockchain nodes/peers may be present in the blockchain network including endorsing peers, which simulate and endorse entries proposed by clients and committing peers which verify endorsements, validate entries, and commit entries to the distributed ledger. In this example, the blockchain nodes may perform the role of endorser node, committer node, or both.

The instant system includes a blockchain that stores immutable, sequenced records in blocks, and a state database (current world state) maintaining a current state of the blockchain. One distributed ledger may exist per channel and each peer maintains its own copy of the distributed ledger for each channel of which they are a member. The instant blockchain is an entry log, structured as hash-linked blocks where each block contains a sequence of N entries. Blocks may include various components such as those shown in FIG. 6D. The linking of the blocks may be generated by adding a hash of a prior block's header within a block header of a current block. In this way, all entries on the blockchain are sequenced and cryptographically linked together preventing tampering with blockchain data without breaking the hash links. Furthermore, because of the links, the latest block in the blockchain represents every entry that has come before it. The instant blockchain may be stored on a peer file system (local or attached storage), which supports an append-only blockchain workload.

The current state of the blockchain and the distributed ledger may be stored in the state database. Here, the current state data represents the latest values for all keys ever included in the chain entry log of the blockchain. Smart contract executable code invocations execute entries against the current state in the state database. To make these smart contract executable code interactions extremely efficient, the latest values of all keys are stored in the state database. The state database may include an indexed view into the entry log of the blockchain, it can therefore be regenerated from the chain at any time. The state database may automatically get recovered (or generated if needed) upon peer startup, before entries are accepted.

Endorsing nodes receive entries from clients and endorse the entry based on simulated results. Endorsing nodes hold smart contracts, which simulate the entry proposals. When an endorsing node endorses an entry, the endorsing nodes creates an entry endorsement, which is a signed response from the endorsing node to the client application indicating the endorsement of the simulated entry. The method of endorsing an entry depends on an endorsement policy that may be specified within smart contract executable code. An example of an endorsement policy is “the majority of endorsing peers must endorse the entry.” Different channels may have different endorsement policies. Endorsed entries are forward by the client application to an ordering service.

The ordering service accepts endorsed entries, orders them into a block, and delivers the blocks to the committing peers. For example, the ordering service may initiate a new block when a threshold of entries has been reached, a timer times out, or another condition. In this example, blockchain node is a committing peer that has received a data block **682A** for storage on the blockchain. The ordering service may be made up of a cluster of orderers. The ordering service does not process entries, smart contracts, or maintain the shared ledger. Rather, the ordering service may accept the endorsed entries and specifies the order in which those entries are committed to the distributed ledger. The architecture of the blockchain network may be designed such that the specific implementation of ‘ordering’ (e.g., Solo, Kafka, BFT, etc.) becomes a pluggable component.

Entries are written to the distributed ledger in a consistent order. The order of entries is established to ensure that the updates to the state database are valid when they are committed to the network. Unlike a cryptocurrency blockchain system (e.g., Bitcoin, etc.) where ordering occurs through the solving of a cryptographic puzzle, or mining, in this example the parties of the distributed ledger may choose the ordering mechanism that best suits that network.

Referring to FIG. 6D, a block **682A** (also referred to as a data block) that is stored on the blockchain and/or the distributed ledger may include multiple data segments such as a block header **684A** to **684n**, transaction-specific data **686A** to **686n**, and block metadata **688A** to **688n**. It should be appreciated that the various depicted blocks and their contents, such as block **682A** and its contents are merely for purposes of an example and are not meant to limit the scope of the example embodiments. In some cases, both the block header **684A** and the block metadata **688A** may be smaller than the transaction-specific data **686A**, which stores entry data; however, this is not a requirement. The block **682A** may store transactional information of N entries (e.g., **100**, **500**, **1000**, **2000**, **3000**, etc.) within the block data **690A** to **690n**. The block **682A** may also include a link to a previous block (e.g., on the blockchain) within the block header **684A**. In particular, the block header **684A** may include a hash of a previous block’s header. The block header **684A**

may also include a unique block number, a hash of the block data **690A** of the current block **682A**, and the like. The block number of the block **682A** may be unique and assigned in an incremental/sequential order starting from zero. The first block in the blockchain may be referred to as a genesis block, which includes information about the blockchain, its members, the data stored therein, etc.

The block data **690A** may store entry information of each entry that is recorded within the block. For example, the entry data may include one or more of a type of the entry, a version, a timestamp, a channel ID of the distributed ledger, an entry ID, an epoch, a payload visibility, a smart contract executable code path (deploy tx), a smart contract executable code name, a smart contract executable code version, input (smart contract executable code and functions), a client (creator) identify such as a public key and certificate, a signature of the client, identities of endorsers, endorser signatures, a proposal hash, smart contract executable code events, response status, namespace, a read set (list of key and version read by the entry, etc.), a write set (list of key and value, etc.), a start key, an end key, a list of keys, a Merkel tree query summary, and the like. The entry data may be stored for each of the N entries.

In some embodiments, the block data **690A** may also store transaction-specific data **686A**, which adds additional information to the hash-linked chain of blocks in the blockchain. Accordingly, the data **686A** can be stored in an immutable log of blocks on the distributed ledger. Some of the benefits of storing such data **686A** are reflected in the various embodiments disclosed and depicted herein. The block metadata **688A** may store multiple fields of metadata (e.g., as a byte array, etc.). Metadata fields may include signature on block creation, a reference to a last configuration block, an entry filter identifying valid and invalid entries within the block, last offset persisted of an ordering service that ordered the block, and the like. The signature, the last configuration block, and the orderer metadata may be added by the ordering service. Meanwhile, a committer of the block (such as a blockchain node) may add validity/invalidity information based on an endorsement policy, verification of read/write sets, and the like. The entry filter may include a byte array of a size equal to the number of entries in the block data **610A** and a validation code identifying whether an entry was valid/invalid.

The other blocks **682B** to **682n** in the blockchain also have headers, files, and values. However, unlike the first block **682A**, each of the headers **684A** to **684n** in the other blocks includes the hash value of an immediately preceding block. The hash value of the immediately preceding block may be just the hash of the header of the previous block or may be the hash value of the entire previous block. By including the hash value of a preceding block in each of the remaining blocks, a trace can be performed from the Nth block back to the genesis block (and the associated original file) on a block-by-block basis, as indicated by arrows **692**, to establish an auditable and immutable chain-of-custody.

The above embodiments may be implemented in hardware, in a computer program executed by a processor, in firmware, or in a combination of the above. A computer program may be embodied on a computer-readable medium, such as a storage medium. For example, a computer program may reside in random access memory (“RAM”), flash memory, read-only memory (“ROM”), erasable programmable read-only memory (“EPROM”), electrically erasable programmable read-only memory (“EEPROM”), registers,

hard disk, a removable disk, a compact disk read-only memory ("CD-ROM"), or any other form of storage medium known in the art.

An exemplary storage medium may be coupled to the processor such that the processor may read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an application-specific integrated circuit ("ASIC"). In the alternative, the processor and the storage medium may reside as discrete components. For example, FIG. 7 illustrates an example computer system architecture 700, which may represent or be integrated in any of the above-described components, etc.

FIG. 7 is not intended to suggest any limitation as to the scope of use or functionality of embodiments of the application described herein. Regardless, the computing node 700 is capable of being implemented and/or performing any of the functionality set forth hereinabove.

In computing node 700 there is a computer system/server 702, which is operational with numerous other general purpose or special purpose computing system environments or configurations. Examples of well-known computing systems, environments, and/or configurations that may be suitable for use with computer system/server 702 include, but are not limited to, personal computer systems, server computer systems, thin clients, thick clients, hand-held or laptop devices, multiprocessor systems, microprocessor-based systems, set-top boxes, programmable consumer electronics, network PCs, minicomputer systems, mainframe computer systems, and distributed cloud computing environments that include any of the above systems or devices, and the like.

Computer system/server 702 may be described in the general context of computer system-executable instructions, such as program modules, being executed by a computer system. Generally, program modules may include routines, programs, objects, components, logic, data structures, and so on that perform particular tasks or implement particular abstract data types. Computer system/server 702 may be practiced in distributed cloud computing environments where tasks are performed by remote processing devices that are linked through a communications network. In a distributed cloud computing environment, program modules may be located in both local and remote computer system storage media including memory storage devices.

As shown in FIG. 7, computer system/server 702 in cloud computing node 700 is shown in the form of a general-purpose computing device. The components of computer system/server 702 may include, but are not limited to, one or more processors or processing units 704, a system memory 706, and a bus that couples various system components including system memory 706 to processor 704.

The bus represents one or more of any of several types of bus structures, including a memory bus or memory controller, a peripheral bus, an accelerated graphics port, and a processor or local bus using any of a variety of bus architectures. By way of example, and not limitation, such architectures include Industry Standard Architecture (ISA) bus, Micro Channel Architecture (MCA) bus, Enhanced ISA (EISA) bus, Video Electronics Standards Association (VESA) local bus, and Peripheral Component Interconnects (PCI) bus.

Computer system/server 702 typically includes a variety of computer system readable media. Such media may be any available media that is accessible by computer system/server 702, and it includes both volatile and non-volatile media, removable and non-removable media. System memory 706,

in one example, implements the flow diagrams of the other figures. The system memory 706 can include computer system readable media in the form of volatile memory, such as random-access memory (RAM) 708 and/or cache memory 710. Computer system/server 702 may further include other removable/non-removable, volatile/non-volatile computer system storage media. By way of example only, memory 706 can be provided for reading from and writing to a non-removable, non-volatile magnetic media (not shown and typically called a "hard drive"). Although not shown, a magnetic disk drive for reading from and writing to a removable, non-volatile magnetic disk (e.g., a "floppy disk"), and an optical disk drive for reading from or writing to a removable, non-volatile optical disk such as a CD-ROM, DVD-ROM or other optical media can be provided. In such instances, each can be connected to the bus by one or more data media interfaces. As will be further depicted and described below, memory 706 may include at least one program product having a set (e.g., at least one) of program modules that are configured to carry out the functions of various embodiments of the application.

Program/utility, having a set (at least one) of program modules, may be stored in memory 706 by way of example, and not limitation, as well as an operating system, one or more application programs, other program modules, and program data. Each of the operating system, one or more application programs, other program modules, and program data or some combination thereof, may include an implementation of a networking environment. Program modules generally carry out the functions and/or methodologies of various embodiments of the application as described herein.

As will be appreciated by one skilled in the art, aspects of the present application may be embodied as a system, method, or computer program product. Accordingly, aspects of the present application may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a "circuit," "module" or "system." Furthermore, aspects of the present application may take the form of a computer program product embodied in one or more computer-readable medium(s) having computer-readable program code embodied thereon.

Computer system/server 702 may also communicate with one or more external devices via an I/O device 712 (such as an I/O adapter), which may include a keyboard, a pointing device, a display, a voice recognition module, etc., one or more devices that enable a user to interact with computer system/server 702, and/or any devices (e.g., network card, modem, etc.) that enable computer system/server 702 to communicate with one or more other computing devices. Such communication can occur via I/O interfaces of the device 712. Still yet, computer system/server 702 can communicate with one or more networks such as a local area network (LAN), a general wide area network (WAN), and/or a public network (e.g., the Internet) via a network adapter. As depicted, device 712 communicates with the other components of computer system/server 702 via a bus. It should be understood that although not shown, other hardware and/or software components could be used in conjunction with computer system/server 702. Examples, include, but are not limited to: microcode, device drivers, redundant processing units, external disk drive arrays, RAID systems, tape drives, and data archival storage systems, etc.

Although an exemplary embodiment of at least one of a system, method, and non-transitory computer-readable

35

medium has been illustrated in the accompanied drawings and described in the foregoing detailed description, it will be understood that the application is not limited to the embodiments disclosed, but is capable of numerous rearrangements, modifications, and substitutions as set forth and defined by the following claims. For example, the capabilities of the system of the various figures can be performed by one or more of the modules or components described herein or in a distributed architecture and may include a transmitter, receiver or pair of both. For example, all or part of the functionality performed by the individual modules, may be performed by one or more of these modules. Further, the functionality described herein may be performed at various times and in relation to various events, internal or external to the modules or components. Also, the information sent between various modules can be sent between the modules via at least one of: a data network, the Internet, a voice network, an Internet Protocol network, a wireless device, a wired device and/or via plurality of protocols. Also, the messages sent or received by any of the modules may be sent or received directly and/or via one or more of the other modules.

One skilled in the art will appreciate that a "system" could be embodied as a personal computer, a server, a console, a personal digital assistant (PDA), a cell phone, a tablet computing device, a smartphone or any other suitable computing device, or combination of devices. Presenting the above-described functions as being performed by a "system" is not intended to limit the scope of the present application in any way but is intended to provide one example of many embodiments. Indeed, methods, systems and apparatuses disclosed herein may be implemented in localized and distributed forms consistent with computing technology.

It should be noted that some of the system features described in this specification have been presented as modules to more particularly emphasize their implementation independence. For example, a module may be implemented as a hardware circuit comprising custom very-large-scale integration (VLSI) circuits or gate arrays, off-the-shelf semiconductors such as logic chips, transistors, or other discrete components. A module may also be implemented in programmable hardware devices such as field-programmable gate arrays, programmable array logic, programmable logic devices, graphics processing units, or the like.

A module may also be at least partially implemented in software for execution by various types of processors. An identified unit of executable code may, for instance, comprise one or more physical or logical blocks of computer instructions that may, for instance, be organized as an object, procedure, or function. Nevertheless, the executables of an identified module need not be physically located together but may comprise disparate instructions stored in different locations that, when joined logically together, comprise the module and achieve the stated purpose for the module. Further, modules may be stored on a computer-readable medium, which may be, for instance, a hard disk drive, flash device, random access memory (RAM), tape, or any other such medium used to store data.

Indeed, a module of executable code could be a single instruction, or many instructions, and may even be distributed over several different code segments, among different programs, and across several memory devices. Similarly, operational data may be identified and illustrated herein within modules and may be embodied in any suitable form and organized within any suitable type of data structure. The operational data may be collected as a single data set or may be distributed over different locations, including over dif-

36

ferent storage devices, and may exist, at least partially, merely as electronic signals on a system or network.

It will be readily understood that the components of the application, as generally described and illustrated in the figures herein, may be arranged and designed in a wide variety of different configurations. Thus, the detailed description of the embodiments is not intended to limit the scope of the application as claimed but is merely representative of selected embodiments of the application.

One having ordinary skill in the art will readily understand that the above may be practiced with steps in a different order and/or with hardware elements in configurations that are different from those which are disclosed. Therefore, although the application has been described based upon these preferred embodiments, it would be apparent to those of skill in the art that certain modifications, variations, and alternative constructions would be apparent.

While preferred embodiments of the present application have been described, it is to be understood that the embodiments described are illustrative only and the scope of the application is to be defined solely by the appended claims when considered with a full range of equivalents and modifications (e.g., protocols, hardware devices, software platforms etc.) thereto

What is claimed is:

1. A method, comprising:

responsive to an inoperability of a vehicle above a threshold, requesting for the vehicle to be remotely operated; receiving software, by the vehicle, based on level of the inoperability;

responsive to an installation of the software on the vehicle, remotely operating the vehicle using the installed software;

responsive to the inoperability of the vehicle above the threshold while it is remotely operated, receiving an update to the software;

responsive to an installation of the updated software on the vehicle, further remotely operating the vehicle using the installed updated software; and

responsive to the inoperability below the threshold, requesting a cessation of the further remote operation of the vehicle, wherein one or more of the software and the updated software assists in maneuvering the vehicle from a scenario where it is difficult to maneuver out of in a time that is less than the threshold.

2. The method of claim 1, wherein one or more of the software and the updated software include an enhancement of functionality to be performed by the vehicle and assists in maneuvering the vehicle.

3. The method of claim 1, responsive to one or more of the remote operating of the vehicle, and the further remote operating of the vehicle, sending data to a server, wherein the data is related to a condition of the vehicle.

4. The method of claim 1, wherein the updated software is based on a new environmental condition of the vehicle after the remote operating, when the new environment is different from an original environmental condition.

5. The method of claim 1, responsive to the inoperability being below the threshold after the updated software is installed on the vehicle, sending a notification of the inoperability to proximate vehicles.

6. The method of claim 1, responsive to the inoperability of the vehicle being above the threshold and a condition of at least one passenger of the vehicle being above another threshold, requesting, by the vehicle, emergency services.

37

7. A system, comprising:
 at least one processor; and
 a memory, wherein the processor and the memory are
 communicably coupled, wherein the processor:
 responsive to an inoperability of a vehicle above a thresh- 5
 old, requests for the vehicle to be remotely operated;
 receives software, by the vehicle, based on level of the
 inoperability;
 responsive to an installation of the software on the
 vehicle, remotely operates the vehicle via the use of the 10
 installed software;
 responsive to the inoperability of the vehicle above the
 threshold, while it is remotely operated, receives an
 update to the software;
 responsive to an installation of the updated software on 15
 the vehicle, further remotely operates the vehicle using
 the installed updated software; and
 responsive to the inoperability below the threshold,
 requests a cessation of the further remote operation of
 the vehicle, wherein one or more of the software and 20
 the updated software assists in maneuvering the vehicle
 from a scenario where it is difficult to maneuver out of
 in a time that is less than the threshold.

8. The system of claim 7, wherein one or more of the
 software and the updated software include an enhancement 25
 of functionality to be performed by the vehicle and assists in
 a maneuver of the vehicle.

9. The system of claim 7, responsive to one or more of the
 remote operating of the vehicle, and the further remote
 operating of the vehicle, the processor sends data to a server, 30
 wherein the data is related to a condition of the vehicle.

10. The system of claim 7, wherein the updated software
 is based on a new environmental condition of the vehicle
 after the remote operation, when the new environment is
 different from an original environmental condition. 35

11. The system of claim 7, responsive to the inoperability
 being below the threshold after the updated software is
 installed on the vehicle, the processor sends a notification of
 the inoperability to proximate vehicles.

12. The system of claim 7, responsive to the inoperability 40
 of the vehicle being above the threshold and a condition of
 at least one passenger of the vehicle being above another
 threshold, the processor requests, by the vehicle, emergency
 services.

38

13. A non-transitory computer-readable storage medium
 comprising instructions, that when read by a processor,
 cause the processor to perform:
 responsive to an inoperability of a vehicle above a thresh-
 old, requesting for the vehicle to be remotely operated;
 receiving software, by the vehicle, based on level of the
 inoperability;
 responsive to an installation of the software on the
 vehicle, remotely operating the vehicle using the 10
 installed software;
 responsive to the inoperability of the vehicle above the
 threshold while it is remotely operated, receiving an
 update to the software;
 responsive to an installation of the updated software on
 the vehicle, further remotely operating the vehicle
 using the installed updated software; and
 responsive to the inoperability below the threshold,
 requesting a cessation of the further remote operation
 of the vehicle, wherein one or more of the software and
 the updated software assists in maneuvering the vehicle
 from a scenario where it is difficult to maneuver out of
 in a time that is less than the threshold.

14. The non-transitory computer-readable storage
 medium of claim 13, wherein one or more of the software 25
 and the updated software include an enhancement of func-
 tionality to be performed by the vehicle and assists in
 maneuvering the vehicle.

15. The non-transitory computer-readable storage
 medium of claim 13, responsive to one or more of the remote
 operating of the vehicle, and the further remote operating of
 the vehicle, sending data to a server, wherein the data is
 related to a condition of the vehicle. 30

16. The non-transitory computer-readable storage
 medium of claim 13, wherein the updated software is based
 on a new environmental condition of the vehicle after the
 remote operating, when the new environment is different
 from an original environmental condition. 35

17. The non-transitory computer-readable storage
 medium of claim 13, responsive to the inoperability of a
 vehicle being above the threshold and a condition of at least
 one passenger of the vehicle being above another threshold,
 requesting, by the vehicle, emergency services. 40

* * * * *